

You work for an online shopping company that allows customers to write product reviews as text on the shopping website and app. You want to enable your developers to extract a user’s sentiment from blocks of text from the user’s review to better understand customer experiences. Which of these technologies would you use for that purpose?

- A) Cloud Translation

✗
- B) Cloud Vision

✗
- C) Cloud Trace

✗
- D) Cloud Natural Language API

✓

Explanation

You would use Cloud Natural Language application programming interface (API) for this scenario. Cloud Natural Language API or Natural Language Artificial Intelligence (AI) enables you to gain understanding of natural language using API. This can help you identify the following aspects in a block of text:

- Key entities
- Positive or negative sentiments
- Language used
- Syntax including dependency trees and speech parts

The Cloud Natural Language API can be used by developers through API calls or through reference to a document that may be stored on Cloud Storage.

Cloud Translation is not the appropriate choice for this scenario. Cloud Translation or Translation AI performs the translation of text between languages using Representational State Transfer (RESTful) APIs. You can use Cloud Translation to translate text from English to Spanish to any other language and vice versa. By using Cloud Translation API, you can provide automatic text translation features for an app or Website.

Cloud Vision is not the appropriate choice for this scenario. Cloud Vision is used by developers to understand image content through machine learning (ML) models accessed through an API. It provides image classification into various categories, object and face detection inside images, and identification of printed words inside of images. Cloud Vision allows you to keep moderation for image content, use analysis of image sentiment for marketing strategies, and create metadata for an image catalog. It also allows you to perform image analysis for images uploaded on requests and can be integrated with Cloud Storage.

Cloud Trace is not the appropriate choice for this scenario. The Cloud Trace service is useful for identifying any bottlenecks that may exist in your system. The latency data collected by Cloud Trace is displayed on the Google Cloud Console. This lets you track the propagation of requests in your application and view near real-time insights of its performance. Cloud Trace can be used to trace performance data from containers, virtual machines (VMs), and projects for App Engine. It also provides automatic alerts when it detects a deviation from the latency profile of your app.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify Google Cloud's solutions for machine learning and AI

References:
[Cloud.google.com > Natural Language AI](#)

Related Material

Your company has a photo sharing service that is heavily in use. You have created an SLO for the service. You need to find a way to check the status of your SLO and also have the error budget computed automatically. Which Google Cloud service would be suitable for this requirement?

- A) Cloud Monitoring

✓
- B) BigQuery

✗
- C) Looker

✗
- D) Cloud Vision

✗

Explanation

Cloud Monitoring can be used to create and use service level objectives (SLOs), service level indicators (SLIs), and error budgets. You can specify an SLO for a month, and Cloud Monitoring computes the corresponding error budget automatically. It can be used to monitor the complete infrastructure of a cloud deployment. It provides a view into the uptime, health, and performance of cloud systems. For this, Cloud Monitoring gathers relevant metadata, events, and metrics from sources that include uptime probes, services, app instrumentation, alerts, notifications, and other components of applications. It allows you to stop issues from happening, and it allows you to identify trends. It also reduces overhead related to monitoring through a service that is integrated and provides metrics, dashboards, and alerts all in one place. Cloud Monitoring provides SLO monitoring and metrics collection for virtual machines (VMs) (using Ops Agent) and Kubernetes (using Managed Service for Prometheus).

You would not use BigQuery for this scenario. BigQuery is a Google service that provides an enterprise-level data warehouse that is fully managed and that can be utilized for analytics. It has features like machine learning (ML), business intelligence, and geospatial analytics ready for use, and has a serverless architecture that allows you to use SQL queries to gain key organizational data. BigQuery has a scalable analysis engine that has a distributed structure. This allows you to perform queries on petabytes of data within minutes.

You would not use Looker for this scenario. Looker is a platform for business intelligence that allows you to access intelligent data for making better business decisions. It also can be embedded into other applications to provide embedded analytics. Looker can be used to create alerts for proactively sending timely insights related to various business data. It also has dashboards that are interactive, which allow for analyzing data in real-time through a trusted data model.

You would not use Cloud Vision for this scenario. Cloud Vision is used by developers to understand image content through machine learning (ML) models accessed through an application programming interface (API). It provides image classification into various categories, object and face detection inside images, and identification of printed words inside of images.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Identify Google Cloud solutions for cloud resource monitoring and application performance management

References:
[Cloud.google.com > Cloud Monitoring](#)

Related Material

A bank has an on-premises data center that it needs to maintain to ensure regulatory compliance. It also has several applications that it uses for internal purposes as well as for serving its customers. What should you use to provide a single portal for visibility of their nodes and clusters in their cloud and on-premises workloads that also manages large volumes of Kubernetes clusters inside their applications?

- A) Firestore

✗
- B) Cloud Functions

✗

C) App Engine

D) Anthos

✖✔

Explanation

You should use Anthos for this scenario. Using Anthos, you can create and manage an application and then run it on both on-premises systems and on the cloud. With Anthos, you can perform migrations straight from virtual machines (VMs), and create and deploy apps using Google Kubernetes Engine (GKE), VMs, and landing zones that are serverless. Anthos allows you to use its platform for deploying both on-premises and cloud-native applications to create a global fleet that has consistency in its operations. Using containers and serverless services, Anthos lets you grow productivity. By maintaining security systems at every stage of the application’s lifecycle, Anthos provides protection for both a software supply chain and application.

You would not use App Engine for this scenario. App Engine allows you to create and host your applications on the same platforms Google uses for several of its own applications. With App Engine, you can build and deploy applications quickly and manage them easily. It provides scalability without the issues related to backups, patching, and hardware provisioning. App Engine provides support for web application development using programming languages like Python, Ruby, C#, Node.js, Go, and PHP.

You would not use Cloud Functions for this scenario. Cloud Functions is best suited for when you need to run functional code based on an event or a trigger. Cloud Functions runs your code when events occur from various sources, such as Pub/Sub, Cloud Storage, and Hypertext Transfer Protocol (HTTP). For example: running a process for video transcoding automatically whenever a video is uploaded to Cloud Storage.

You would not use Firestore for this scenario. Firestore is a serverless document database that is scalable and fully managed. It enables fast development for applications for the internet, mobile, and the Internet of Things (IoT), which can connect to the database directly. Firestore can be used to create real-time applications by using its native synchronization features. It also allows you to customize various rules for security and data validation to protect your data. Firestore can be integrated with BigQuery and Firebase. Firestore is an upgrade to the older service called Datastore, which provides a NoSQL database that is scalable.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Understand modernizing applications with Google Cloud

References:
[Cloud.google.com > Anthos](#)

Related Material

Question #4 of 60

Question ID: 1495197

A company that develops database applications is looking for a Google Cloud service that will enable its applications to get better performance by incorporating the availability and scalability of Redis services without the associated complexities of managing the Redis deployments. Which service should you use in this scenario?

A) Memorystore

B) Datastore

C) Cloud Bigtable

D) Firestore

✔✖✖✖

Explanation

You should use Memorystore in this scenario. Memorystore for Redis is a scalable, fully managed service for Redis on Google Cloud that provides 99.9% availability using up to five read replicas that are distributed across multiple availability zones (AZs). Memorystore provides automation for complex Redis and Memcached tasks, which include ensuring high availability, creating failovers, and performing patching. It is compatible with the protocols of both Memcached and Redis. Memorystore also has provision for private IP addresses and virtual private cloud networks (VPCs) for data security and has integration with Identity and Access Management (IAM) for access control. Memcached provides in-memory caching for temporary data storage. Redis is a key-value NoSQL database using an in-memory data system for high performance. Redis supports sharding and uses persistent data, while Memcached does not.

Firestore is not the appropriate choice for this scenario. Firestore is a serverless document database that is scalable and fully managed. It enables fast development for applications for the Internet, mobile, and the Internet of Things (IoT) that can connect to the database directly. Firestore can be used to create real-time applications by using its native synchronization features. It also allows you to customize various rules for security and data validation to protect your data. Firestore can be integrated with BigQuery and Firebase. Firestore is an upgrade to the older service called Datastore, which provides a NoSQL database that is scalable.

Dataprep is not the appropriate choice for this scenario. Dataprep is a data service for exploring, cleaning, and preparing data visually for the purpose of machine learning (ML) and analysis. It is an intelligent data service that is serverless, so the user does not need to deploy or manage the infrastructure. Dataprep suggests the data transformations that are ideal each time a User Interface (UI) input is detected, so the user does not need to write code for these applications. Dataprep also detects any anomalies that may exist in the data, including duplicates and missing value data. It helps the user avoid having to assess the quality of the data and proceed directly to exploratory data analysis.

Cloud Bigtable is not the appropriate choice for this scenario. Cloud Bigtable is a NoSQL database service that is fully managed and enables you to work with massive workloads for analytics and operations with availability at 99.999%. It is ideal for uses that include applications for IoT, finance, banking, personalized services, and advertising technology. Cloud Bigtable has a storage engine that allows for improved predictions in applications utilizing ML. It also has integration with tools for big data including Hadoop, Dataproc, and Dataflow.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify common Google Cloud solutions for smart analytics

References:
[Cloud.google.com > Memorystore](#)

Related Material

Question #5 of 60

Question ID: 1497160

Your company uses Google’s GKE for automatically managing and deploying container-based applications. As per the shared responsibility model, which of these is the responsibility of the customer?

A) Workloads containing docker files

B) The control pane

C) The operating system of nodes

D) Encryption of data at rest

✔✖✖✖

Explanation

The customer is responsible for workloads that contain docker files, application code, container images, Identity and Access Management (IAM) and Role-Based Access Control (RBAC) policies, and pods. The customer is also responsible for the nodes that are running workloads. This also includes the software that may be installed on nodes as well as any changes to the configuration of the nodes. Node updating is also the responsibility of the customer.

The operating system of nodes, encryption of data at rest, and the control pane are the responsibility of Google. This also includes the cloud infrastructure, which comprises OS, networking, storage, firmware, and hardware. Google is also responsible for the encryption of data at rest and in transit. The operating system in use on nodes, which can be Ubuntu or Container-Optimized OS (COS). is the responsibility of Google. Google Kubernetes Engine (GKE) can automatically deploy upgrades to the images. This is separate from the OS that is running inside of containers. Also, GKE manages the control pane that comprises master virtual machines (VMs), IAM authentication systems, cluster certificate authority (CA), and the application programming interface (API) server. The necessary patching and upgrades needed for these components are supported by a service level objective (SLO). The control pane is the part of the Kubernetes system that controls how communication with a cluster is performed and how user states are implemented. Google hardens this control pane to protect against vulnerabilities.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Describe a cloud security approach and Google Cloud security benefits

References:
[Cloud.google.com > Exploring container security: the shared responsibility model in GKE](#)

Related Material

Question #6 of 60

Question ID: 1495196

You have several teams working on creating applications, and you need to design a system that enables the applications made by each team to interact with each other asynchronously. Which of these services should you utilize for this scenario?

- A) BigQuery✖
- B) Pub/Sub✔
- C) Dataproc✖
- D) Dataprep✖

Explanation

You should utilize Pub/Sub for this scenario. Pub/Sub is a middleware for messaging that enables applications to send messages to each other asynchronously. An application designated as a “publisher” application sends messages to a topic, and all the applications that have subscribed to that topic will receive the messages. This way, senders are decoupled from receivers in an architecture that is referred to as a “loosely-coupled” architecture, which forms the basis of a microservices-based architecture. Pub/Sub is useful for application developers that need to create a communication system between applications that are built independently of each other.

You would not utilize Dataproc in this scenario. Dataproc is better suited for applications related to data lakes. It provides a fully managed service that is scalable and allows you to run Presto, Apache Flink, and Apache Spark frameworks. You can perform data lake modernization, Extract, Transform, and Load (ETL) operations, and provide security for data science using Dataproc. It uses Apache Ranger and Kerberos for configuring security.

You would not utilize BigQuery for this scenario. BigQuery is a Google service that provides an enterprise-level data warehouse that is fully managed and that can be utilized for analytics. It has features like machine learning (ML), business intelligence (BI), and geospatial analytics ready for use, and has a serverless architecture that allows you to use SQL queries for gaining key organizational data. BigQuery has a scalable analysis engine that has a distributed structure. It allows you to perform queries on petabytes of data within minutes.

You would not utilize Dataprep for this scenario. Dataprep is a data service for exploring, cleaning, and preparing data visually for the purpose of ML and analysis. It is an intelligent data service that is serverless, so the user does not need to deploy or manage the infrastructure. Dataprep suggests the data transformations that are ideal each time a User Interface (UI) input is detected, so the user does not need to write code for these applications. Dataprep also detects any anomalies that may exist in the data, including duplicates and missing value data. It helps the user avoid having to assess the quality of the data and proceed directly to exploratory data analysis.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify common Google Cloud solutions for smart analytics

References:
[Cloud.google.com > Pub/Sub](#)

Related Material

Question #7 of 60

Question ID: 1495254

You work for a digital publishing company that owns millions of digital assets, including images, videos, and music. You want to find a way to monetize these digital assets to grow your business value. Which of these Google Cloud services would help you achieve this goal?

- A) GKE✖
- B) Payment Gateway✖
- C) Dataflow✖
- D) Apigee✔

Explanation

You should use Apigee to help you achieve this goal. With Apigee, you can increase the value of your digital assets by monetizing your application programming interfaces (API)-based products. Apigee is used for building, securing, scaling, and analyzing APIs at any location with support for the complete API lifecycle. It provides strong systems of API control and visibility. Apigee lets you monitor APIs using artificial intelligence (AI) powered systems. This ensures your APIs are delivering expected performance and maintaining high availability for customers and developers. You can also identify any API issues and diagnose them easily using the monitoring systems for Apigee. Additionally, Apigee allows you to use AI with your APIs historical metadata and perform behavior prediction as well as anomaly detection in real-time. Apigee also helps maintain regulatory compliance.

You would not use Payment Gateway for this scenario. With Payment Gateway, you can perform processing of payments that are API-based. It can scale up to cater to unpredictable surges in usage as well as be extended to third-party insights and anomaly finding tools through APIs. It also allows for payment processing from across multiple regions of Google Cloud's global infrastructure. Payment Gateway also has built-in tools that allow you to monitor, debug, and trace API-based purchases with high availability and less overhead. You can also export data of transactions into a project on Google Cloud using an export API.

You would not use Google Kubernetes Engine (GKE) for this scenario. GKE is not used for digital asset monetization, but it allows you to create and manage Kubernetes clusters. Kubernetes provides an open-source system to automatically manage, deploy, and scale container-based applications. GKE provides detailed control over containers and nodes, enabling you to specify CPU, RAM, storage, GPUs, and network configurations along with an Autopilot option for situations where there is not enough operational and developer support available.

You would not use Dataflow for this scenario. Dataflow provides an economical, swift, and serverless system for stream and batch processing work. It is a fully managed service that automates the process of provisioning and managing processing resources. It provides the highest possible resource utilization by performing horizontal autoscaling for various worker resources.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Understand the value of APIs

References:
[Cloud.google.com > Apigee API Management](#)

Related Material

Question #8 of 60

Question ID: 1497158

Your company is a data processing company that has teams around the world. You need to optimize the costs of your cloud deployment while making datasets available globally for faster access for teams. Which of these cost management strategies should you use for the business?

- A)** Configure a lifecycle policy for data objects
- B)** Create multiple copies of data in several buckets
- C)** Set auto stop for VM instances
- D) Use multi-region buckets for data storage**
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Explanation

You should use multi-region buckets for data storage. By using multi-region buckets, you can provide greater resiliency to teams distributed globally. This also removes the need for creating duplicate data, which is not a cost optimized approach. Multi-region buckets provide asynchronous replication across multiple geographic locations and also provide backup against disasters and regional outages.

You would not create multiple copies of data in several buckets in this scenario. This will cause data duplication and is not a recommended practice. Duplicating data causes a waste of storage and capital. Some ways you can prevent creating duplicate data is by using multi-region buckets and setting object versioning on Cloud Storage correctly. By specifying an object versioning policy, you can maintain a suitable number of copies of data, and you can also use the bucket lock version for protection from the accidental deletion of objects.

You would not set auto stop for virtual machine (VM) instances in this scenario. This is not related to cost optimization for storage. Using auto start and stop for VMs can help you reduce operating costs for Compute Engine. By deactivating VMs during non-business hours, you can make significant savings on costs rather than allowing VMs to run continuously. You can use a serverless system to automate the process of VM shutdowns by using Cloud Pub/Sub.

You would not configure a lifecycle policy for data objects in this scenario. This does not stop the duplication of data. Google Cloud Storage offers various storage classes, which include standard, nearline, coldline, and archival. By moving data to less expensive storage classes, such as the archival class, you can save on costs for storing data that is accessed rarely. By using a lifecycle policy, you can automatically ensure that data objects change their storage class based on conditions you specify, which can be the age of the object.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Describe financial governance in the cloud and Google Cloud's recommended best practices for effective cloud cost management

References:
[Cloud.google.com > Best practices for optimizing your cloud costs](#)

Related Material

Question #9 of 60

Question ID: 1497127

You need to recommend a Google product to a company that wants to enhance its applications by incorporating intelligent components that are plug-and-play. For this, the company needs to create, deploy, and scale its machine learning models quickly using a single platform. Which Google product should you recommend for this scenario?

- A)** Cloud Life Sciences
- B)** Memorystore
- C) Network Intelligence Center**
- D)** Vertex AI
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Explanation

You should recommend Vertex Artificial Intelligence (AI) for this scenario. Vertex AI provides a singular application programming interface (API) platform for building and scaling machine learning (ML) models swiftly with features for custom tooling and pre-training. It allows you to manage the AI and ML development lifecycle for a company. Vertex AI has various services and tools within it, such as Automated Machine Learning (AutoML), which is a system that allows you to perform training for ML models without the need for coding. Vertex AI also allows you to create ML models that are advanced using its custom model tooling. It has Machine Learning Operations (MLOps) tools that enable ML model maintenance using the Vertex AI Pipelines, which streamlines the process of running pipelines for ML, and the Vertex Feature Store, which you can use to serve and use AI technologies as features for your ML applications. ML is the science of developing intelligent systems and applications that can “learn” from existing data provided to it and it can then provide services that are intuitive, dynamic, and adaptive. ML is a subset of AI that involves methods for making a machine mimic the ability of a human to make decisions and solve problems. This is done using deep learning and neural networks that enable the ML system to learn continuously. It removes the need to program the ML system explicitly by providing it with large quantities of data.

You would not recommend Network Intelligence Center for this scenario. Network Intelligence Center is used to monitor, verify, and optimize a network. It can be used for Google Cloud, on-premises, and multi-cloud systems. Network Intelligence Center reduces the risk of outages occurring and ensures greater levels of compliance with regulations and security.

You would not recommend Memorystore for this scenario. Memorystore is a fully managed service that provides in-memory storage of data with less than a millisecond of response time to access the data. It does this by deploying distributed caches for applications. Memorystore helps migrate the cache layer to the cloud without needing to modify existing application code.

You would not recommend Cloud Life Sciences for this scenario. Cloud Life Sciences enables you to manage, process, and transform data related to life sciences. This data can include genomics and biomedical data. Cloud Life Sciences provides workflows that are containerized and allow for the processing, analysis, and annotations of life sciences data with scalability.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify Google Cloud's solutions for machine learning and AI

References:
[Cloud.google.com > Vertex AI](#)

Related Material

Question #10 of 60

Question ID: 1495288

Your operations team spends a lot of time on repetitive tasks. You want to bring greater efficiency into your Google Cloud operations by reducing toil. What would be the best strategy to achieve this goal?

- A) Use more observability tools**
- B) Perform encryption of data at rest**
- C) Create new SLAs with users**
- D) Halt deployments**
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Explanation

You should use more observability tools to reduce toil. Toil comprises manual and repetitive tasks, which should be automated so that team members can focus their time on higher value work. By using observability tools like Cloud Tracing, Cloud Monitoring, and Cloud Logging, you can gain more insight into services and ensure their correct automation. Creating more automation of services can help reduce toil in an organization, which boosts work efficiency and productivity. A Site Reliability Engineering (SRE) recommended best practice is for operations teams to look into issues with services and attempt to identify the reasons for failure. For this, the teams should gain more knowledge on how a service operates.

You would not perform encryption of data at rest to reduce toil. Encryption at rest of stored data is a way of implementing security for data stored on the cloud infrastructure. Google utilizes multiple layers of encryption for the protection of data at rest. Google's storage infrastructure encrypts all user data by default before it is stored on physical storage. This way, encryption is performed by Google Cloud at the storage or application layer of the cloud infrastructure.

You would not create new service level agreements (SLAs) with users to reduce toil. SLAs are an agreement between the service provider and customer that specifies the level of availability that the service will have. Reduction of toil involves creating more automation of systems, which removes the need for manual involvement of team members.

You would not halt deployments to reduce toil. Halting deployment is not done for toil reduction but is done in situations where an error budget is consumed. When new features are added to a service, new bugs may be introduced, which can eat into an existing error budget. Once an organization reaches a stage where the error budget for a given period is exhausted, then management will need to halt the development of new features until the error budget is restored.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Identify Google Cloud solutions for cloud resource monitoring and application performance management

References:
[Services.google.com > Increasing Business Value with Better IT Operations: A guide to Site Reliability Engineering \(SRE\)](#)

Related Material

Question #11 of 60

Question ID: 1495239

Your company is a software development firm that has been using a legacy on-premises system and now wants to leverage cloud technology to reduce bugs caused by differences in the development and production environments. What kind of change pattern for application modernization should you opt for in this use case?

- A) Perform containerization of all services gradually**
- B) Keep specific personnel dedicated to testing tasks**
- C) Combine multiple services into a single monolith**
- D) Use a manual release process for all systems**
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Explanation

You should perform containerization of all services gradually. When a development environment is too different from an existing production environment, there can be various bugs and errors generated. A solution to this is to containerize all services gradually over time. This creates a greater level of control and predictability over the environments for development, production, and staging. You start by containerizing a single service. After you have gained confidence using certain processes and tools for containerization, you can proceed to containerize all services. This is a key aspect of application modernization.

You would not use a manual release process for all systems. A release process for software that is performed manually is prone to errors and is slower than an automated system. A key DevOps practice is continuous integration and continuous delivery (CI/CD), which specifies that a controlled and systematic approach needs to be used when making changes to ensure system stability and avoiding configuration drift. A public cloud system offers features like CI/CD for rapid and automated development and updates for applications. You would first use a simple CI model and then employ CD. After some time, you would apply the CI/CD model to all existing services. Then you can add additional stages to your CI/CD pipeline gradually.

You would not keep specific personnel dedicated to testing tasks. You would instead opt for automated testing systems. Manual processes for testing are prone to human errors. For this reason, when performing application modernization, an automated approach is recommended. By utilizing a basic CI/CD pipeline for your containers, you can have an automated system for testing and deploying your software.

You would not combine multiple services into a single monolith. Decoupling is a key part of modernizing applications that allows for greater flexibility in making changes to parts of the application independently. This is a form of functional decomposition that breaks down an existing legacy or “monolith” application into independent applications. One way to establish decoupling is using services like Pub/Sub. Pub/Sub is a middleware for messaging that enables applications to send messages to each other asynchronously. An application designated as a “publisher” application sends messages to a topic, and all the applications that have subscribed to that topic will receive the messages. This way, senders are decoupled from receivers in an architecture that is referred to as a “loosely-coupled” architecture, which forms the basis of a microservices-based architecture.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Understand modernizing applications with Google Cloud

References:
[Inthecloud.withgoogle.com > CIO Guide to Application Modernization](#)

Related Material

Question #12 of 60

Question ID: 1495220

Your company currently has a legacy on-premises data center, which it uses to provide online shopping services to its customers. Which of these features will require the company to modernize its IT infrastructure with Google Cloud?

- A) Increasing the physical security of the data center**
- B) Deploying global applications that are fault tolerant and highly available**
- C) Connecting legacy systems to the Internet using VPNs**
- D) Performing vertical scaling of data center infrastructure**

Explanation

Deploying global applications that are fault tolerant and highly available requires modernizing IT infrastructure with Google Cloud. A legacy infrastructure cannot deliver a global fault tolerant infrastructure at the level and scalability of a cloud deployment. Google Cloud provides a global infrastructure that consists of regions and zones. This gives you the flexibility to choose how you want to deploy your applications for the best levels of reliability and availability. It also provides the least latency possible for end users through the use of edge locations. Regions are geographical areas that are independent. Each region consists of zones, which are areas of deployment of Google resources inside of a region. A zone can be considered as a single failure domain, which implies that for an application to have fault tolerance and high availability, it needs to be deployed across multiple zones. The services of Google's infrastructure are available globally and have round the clock maintenance and upgrades performed. Google’s global infrastructure makes the process of managing multi-region deployments simple and provides flexible load balancing features.

Performing vertical scaling of data center infrastructure does not require a cloud deployment. Vertical scaling involves increasing the computing power of existing machines and can be done within a data center itself. This is not a cloud exclusive feature. However, a cloud deployment would be more appropriate if there was a requirement for computing resources to be adjusted dynamically as per changing user demand. This is a concept called elasticity, which ensures that the optimum amount of cloud resources is allocated for handling workloads.

Connecting legacy systems to the Internet using virtual private networks (VPNs) does not require a cloud deployment. For connecting computer systems to the Internet, you need an Internet connection through an Internet Service Provider (ISP), and a VPN connection through a VPN provider. However, modernizing the infrastructure with Google Cloud does have the advantage of improving the security of the existing systems through strong identity and access management as well as network protection tools to safeguard against Internet-based malware.

Increasing the physical security of the data center does not require a cloud deployment. The physical security of the data center is the responsibility of the company and needs to be done by the company itself. The security measures will include security barriers, such as walls and fencing; and monitoring systems, such as cameras and sensors. Security measures might also include guards and card- or token-based access control.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Learn what modernizing IT infrastructure with Google Cloud means

References:
[Cloud.google.com > Geography and regions](#)

Related Material

Question #13 of 60

Question ID: 1497142

Your company has built and deployed various machine learning models using Vertex AI and wants to find ways to use these models to make predictions. How would you serve predictions for models deployed using Vertex AI?

- A) Using endpoints**
- B) Using an OLTP database**
- C) By extraction of structured data from documents**
- D) Through data labelling jobs**

Explanation

You can serve predictions for models deployed using Vertex Artificial Intelligence (AI) using endpoints. You can use endpoints even for models that were not trained on Vertex AI. Vertex AI provides a singular application programming interface (API) platform to build and scale machine learning (ML) models swiftly with features for custom tooling and pre-training. It allows you to manage the AI and ML development lifecycle for a company. Vertex AI has various services and tools within it, like Automated Machine Learning (AutoML), which is a system that allows you to perform training for ML models without the need for coding. Vertex AI also allows you to create ML models that are advanced using its custom model tooling. It also has Machine Learning Operations (MLOps) tools that enable ML model maintenance using the Vertex AI Pipelines, which streamlines the process of running pipelines for ML and the Vertex Feature Store that you can use for serving and using AI technologies as features for your ML applications.

You would not serve predictions for models deployed using Vertex AI through data labeling jobs. You use data labeling jobs for requesting human labeling for existing datasets. These datasets can be used for training an ML model that is customized.

You would not serve predictions for models deployed using Vertex AI by using an online transaction processing (OLTP) database. You would instead use Cloud Spanner to provision a highly scalable database for OLTP applications. This database will have high levels of availability and consistency worldwide.

You would not serve predictions for models deployed using Vertex AI by extraction of structured data from documents. You can use Document Artificial Intelligence (AI) for the classification and extraction of structured data from existing documents. This can enable you to find key insights and perform automation of organizational practices.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify Google Cloud's solutions for machine learning and AI

References:
[Cloud.google.com > Introduction to Vertex AI](#)

Related Material

Question #14 of 60

Question ID: 1495222

The company you work for wants to find more secure and cost-effective methods of managing its systems. You are looking to select the appropriate cloud deployment for this. Which of these deployment configurations uses a hybrid cloud infrastructure?

- A)** Using SaaS from one cloud provider and PaaS from another
- B)** Storing all organizational data and applications in an on-premises data center
- C) Using an on-premises data center for storing raw data along with a public cloud for data analytics**
- D)** Using one public cloud service provider for a web frontend and a different one for the backend

Explanation

Using an on-premises data center for storing raw data along with a public cloud for data analytics is an example of a hybrid cloud infrastructure. In a hybrid cloud system, a company uses its own data center along with a public cloud system to achieve its business goals. A hybrid cloud also always contains a private cloud. A hybrid cloud deployment allows a company to run its applications on-premises and use the public cloud for bursting when user demands go beyond the limits of the data center.

Using one public cloud service provider (CSP) for a web frontend and a different one for the backend is not an example of a hybrid cloud deployment but of a multi-cloud deployment. A multi-cloud deployment involves using multiple public CSPs where each public cloud services a different function for the company. A company can then use one public cloud for database services and another one for analytics. This lets a company use its preferred cloud provider for specific services. Remember that using multiple public cloud service providers indicates a multi-cloud infrastructure.

Using Software as a Service (SaaS) from one cloud provider and Platform as a Service (PaaS) from another is not an example of a hybrid cloud infrastructure but of a multi-cloud infrastructure. This is because both SaaS and PaaS are public cloud offerings, and using several public clouds is a feature of multi-cloud deployments. The PaaS model allows consumers to install their own applications onto a cloud platform. The provider supplies the necessary operating system and physical hardware. This allows consumers to quickly deploy their own applications without the burden of first acquiring the necessary physical resources and operating system(s) to make it happen. The SaaS model covers the use of software applications by consumers who are running on a provider's cloud infrastructure. These applications can be accessed using thin client interfaces, including mobile devices and web browsers. An example of this is email accessible from a web browser, such as Gmail. The advantage of this model is that a consumer can utilize the application without the need to physically install the application or be concerned about the infrastructure needed to run it.

Storing all organizational data and applications in an on-premises data center is not an example of a hybrid cloud infrastructure but of a private cloud infrastructure. In this infrastructure, all the infrastructure and applications that are part of the cloud or virtualization system are running on systems that are inside a company's on-premise data center.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Learn what modernizing IT infrastructure with Google Cloud means

References:
[Cloud.google.com > Hybrid and multi-cloud architecture patterns](#)

Related Material

Question #15 of 60

Question ID: 1495898

Your company is considering using a hybrid cloud setup where you will use several public clouds and your existing on-premises servers.

Which cloud migration method would you use if you need to refactor an existing application and modernize it before moving it to a cloud deployment?

- A) Lift, tinker, and shift**
- B)** Rip and replace
- C)** Lift and shift
- D)** Improve and move

Explanation

You would use the “Improve and move” method in this scenario. This method involves taking an existing application and first refactoring and modernizing it. Then the application is moved to a public cloud system. This movement to a public cloud is an ideal situation for companies that want to create new products and services using new technologies.

You would not use the “Lift and shift” method in this scenario. This method involves the migration of an application that was previously in a private cloud environment to a public cloud without making any key modifications. This approach is used by companies that want minimal disruption to their operations, and want to migrate to the cloud while keeping their operating costs as low as possible.

You would not use the “Rip and replace” method in this scenario. This method involves replacing an existing application completely with a new cloud-based version. This approach is used for situations where modernizing the existing application is simply not a feasible alternative due to cost or technology related limitations.

You would not use the “Lift, tinker, and shift” method in this scenario. This method is for situations where you need to modify parts of an application to gain the benefits of a cloud deployment without changing the underlying core of the application.

When considering migration strategies for a hybrid cloud deployment, you also need to be aware of the utilities of a hybrid cloud system:

- Well-paced modernization – A hybrid cloud approach allows you to migrate your applications to a cloud at a speed that is comfortable for your organization. This way you can change the IT infrastructure of your organization over a period of time.
- Regulatory compliance – Each industry has certain stipulations that specify where an application is allowed to operate. Using a hybrid cloud deployment ensures that you keep the use of your application and data within regulatory compliance laws.
- On-premises applications – Some of your applications may need to be run on mainframe computers on-premises. Moving these systems to a cloud deployment may not be feasible. For such regulated applications, a hybrid cloud deployment may be suitable.
- Applications at edge locations – Some industries and applications require the use of edge computing to provide low latency. This could be for systems like retail kiosks or telecommunication networks. Hybrid cloud systems allow you to run such specific applications at remotely placed edge locations.

Objective:
Introduction to Digital Transformation with Google Cloud

Sub-Objective:
Explain why cloud technology is revolutionizing business

References:
[Cloud.google.com > What is a Hybrid Cloud? > How-to hybrid](#)

Related Material

Question #16 of 60

Question ID: 1495183

Your company is creating a new e-commerce portal and wants to begin the process of data classification and management for it. You need to identify all the possible structured and unstructured data you will have to work with. Which of these is an example of structured data?

- A) Customer ratings for each product**
- B) Pictures of products featured on the site**
- C) Descriptions of the products written by vendors**
- D) Reviews written for products by customers**
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Explanation

Customer ratings for each product are an example of structured data. This is because a product rating can be a numerical value or score. Credit card numbers and product names are also examples of structured data. Structured data is data that is organized and quantitative. This allows for easy storage and management using a database. Databases provide an organized way of storing data as tables that can be accessed using a computing system, and can ingest large amounts of data from various sources. The consistency and accuracy levels of data stored in a database comprise the transaction integrity.

The other options are not examples of structured data. They are unstructured data. Product reviews and descriptions both comprise text files, and product pictures are image files. Image files and documents from word processing applications are examples of unstructured data. Unstructured data comprises data that is qualitative and is stored as objects. This allows the data to be stored in its original format along with key metadata that provides ways to uniquely identify the data. Metadata is data about stored objects.

Some additional terms you need to know include:

- Data point – This represents a chunk of information, which can be customer purchases.
- Dataset – These are an aggregation of several data points, which belong to a particular category like “transactions by customers”.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify common Google Cloud solutions for data management

References:
[Services.google.com > Cloud Digital Leader Study Guide](#)

Related Material

Question #17 of 60

Question ID: 1495193

A company that is in the digital publishing domain has approached you to help them find a suitable technology for building a data warehouse that they can use for performing analysis and reporting with scalability that is seamless. Which Google Cloud service should you recommend for this requirement?

- A) DataFlow**
- B) Cloud Trace**
- C) BigQuery**
- D) Cloud Storage**
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Explanation

You should recommend BigQuery for this scenario. BigQuery is a Google service that provides an enterprise-level data warehouse that is fully managed and that can be utilized for analytics. It has features like machine learning (ML), business intelligence (BI), and geospatial analytics ready for use and has a serverless architecture allowing you to use SQL queries for gaining key organizational data. BigQuery has a scalable analysis engine that has a distributed structure. This allows you to perform queries on petabytes of data within minutes.

You would not recommend DataFlow for this scenario because it is not a data warehousing service but a service for performing data processing. DataFlow is a fully managed service that automates the process of provisioning and managing processing resources. It provides the highest possible resource utilization by performing horizontal autoscaling for various worker resources.

You would not recommend Cloud Storage for this scenario because it is an object storage service and not a data warehousing service. Cloud Storage can be used for Object Lifecycle Management (OLM), where data can be configured so that data is automatically moved to lower-cost storage classes when the data meets specified criteria like a period having passed or a new version of the data created. Cloud Storage also provides data redundancy through various locations for storage buckets. This provides fast response times and disaster recovery (DR) options.

You would not recommend Cloud Trace in this scenario as it is used as a system for tracing that gathers data related to latency from various applications. The Cloud Trace service is useful for identifying any bottlenecks that may exist in your system. The latency data collected by Cloud Trace is displayed on the Google Cloud Console. This lets you track the propagation of requests in your application and view near real-time insights of its performance. Cloud Trace can be used to trace performance data from containers, virtual machines (VMs), and projects for App Engine. It also provides automatic alerts when it detects a deviation from the latency profile of your app.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify common Google Cloud solutions for smart analytics

References:
[Cloud.google.com > What is BigQuery?](#)

Related Material

Question #18 of 60

Question ID: 1495189

A company currently uses its on-premises data center for data storage, and they have a varying number of users accessing their website each month. The company is considering moving to a cloud deployment. Which of these advantages can they expect to gain from this decision?

A) Source code debugged by the cloud service provider

B) Scalability for storage that is agile

C) Service availability at 100%

D) Adoption of a capital expenditure system

Explanation

The company can expect scalability for storage that is agile. This is a key advantage of using cloud technology as it ensures that the company will only need to pay for resources that they actually consume. Additionally, due to the scalability of the cloud, the company can avail of the benefit of elasticity, where resources can be automatically added or removed to match the fluctuation in the amount of website usage.

The company cannot expect service availability at 100%. This is because a realistic measure of service availability is the five 9s uptime, which corresponds to services available 99.999% of the time. This can indicate downtime of under 6 minutes per year. Agreements related to system uptime are done using a service level agreement (SLA) between a cloud service provider (CSP) and the customer. The SLA specifically provides a formal definition of uptime. An uptime measure of five 9s indicates a highly available system. As an example, the Google Compute Engine SLA specifies that instances in multiple zones will have a minimum monthly uptime of 99.99%. Google Cloud Storage offers 99.999999999% or eleven 9s of annual durability.

The company cannot expect source code debugged by the CSP. This is because even when a company's infrastructure moves from an on-premises data center to a cloud deployment, the company's development team will still be fully responsible for the source code. Cloud deployments follow a shared responsibility model where the CSP is responsible for the cloud infrastructure and its security, and the company is responsible for the data that is stored and used in the cloud.

The company cannot expect the adoption of a capital expenditure system. This is because when a company moves to a cloud deployment, it essentially moves from the existing capital expenditure system to an operational expenditure system. The key difference between the two is that in a capital expenditure system, the company will take on the costs of procuring all the infrastructure and resources they need for running its IT systems. In this system, underutilization of resources and scalability issues can lead to wasted capital for the company. To counter this, an operational expense model allows a company to pay for resources consumed for its operations dynamically. This is also characterized by the pay-as-you-go (PAYG) model, where a cloud customer only pays for resources consumed.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify common Google Cloud solutions for smart analytics

References:
[Cloud.google.com > Compute Engine Service Level Agreement \(SLA\)](#)

Related Material

Question #19 of 60

Question ID: 1497161

A company has a sketching app used by millions of users around the world, and now it needs to make some changes to the application's front end. Which would be the recommended course of action as per Site Reliability Engineering (SRE) best practices?

A) Release the modification to a small subset of users for testing

B) Release the modification to all users to ensure coverage

C) Do not make the release until automation for toil is removed

D) Do not make the release until there is a zero chance of errors

Explanation

As per Site Reliability Engineering (SRE) best practices, it is recommended that the company release the modification to a small subset of users for testing. A key SRE best practice is ensuring that changes are released gradually with testing performed for the changes. Some SRE best practices for production servers include:

- Control failures – Ensure failures don't impact the functionality and allow systems to work using a known configuration and only use new data or configurations after they have been sanitized and validated by a human user.
- Make rollouts progressive – All rollouts and updates that need to be performed should be performed in stages and monitored throughout. The changes should be applied to a small section of traffic across time frames and geographies. This allows for problems to be detected and mitigated.
- Specify service level objectives (SLOs) from a user’s perspective – This involves setting measures for performance and availability using metrics that are applicable to a user. An example of this is the latency and error rates of a client application.
- Maintain a balance between error budgets and innovation – The system’s reliability and speed of performing innovation need to be in balance with error budgets, which are the acceptable failure levels of a specified service. An example is a 0.01% error budget for a service with 99.99% availability. This implies that a team can plan to add new features and innovation to a service only if the service has not already availed its error budget for the month, which includes errors and downtime.
- Performing monitoring – System monitoring for health can yield outputs that include pages, tickets, and logging, and require human intervention depending on the severity of a reported issue.
- Do blameless postmortems – This ensures that systems are improved in their design so that complete groups of possible errors that could arise are mitigated. It is done by not blaming people and focusing on making changes to technology and processes.
- Plan for capacity – This includes making plans to handle planned and unplanned outages while ensuring it does not affect a user beyond an acceptable limit. It involves performing load testing to find resource-to-capacity relationships and validating demand forecasts with actual demands until the two match consistently.
- Checking for overloads – This involves making sure that services still produce acceptable results even if overloaded so that users can continue using them. It may require certain features of a service to be halted in an overload situation, while the main service features are still made available.
- Limit operational tasks for SRE teams – This is to make sure that SRE teams do not dedicate more than half of their time to performing operational tasks (toil) that should be sent to development teams. For this, you can ensure there are regular meetings between SRE and development teams. Toil comprises work that may be manual, nonenduring, automatable, repetitive, and tactical.

You must not release the modification to all users to ensure coverage. As per SRE best practices, until a rollout or modification is tested and verified by a small set of users across various geographies and time frames, it must not be made available to the full set of users. This is to ensure any bugs or errors are mitigated before the change reaches all users,

Not making the release until there is a zero chance of errors is not a part of SRE best practices. Errors and failures are a natural part of the SRE process, but failure needs to be dealt with in a controlled manner. As per SRE best practices, you need to ensure that a steady state of the service is maintained so that any failure that occurs is contained and does not cause a catastrophic failure of the entire service.

Not making the release until automation for toil is removed is not an SRE best practice. On the contrary, a key goal of SRE is to ensure toil automation takes place. Toil automation removes the need for performing manual, repetitive work, which does not have an enduring value.

Objective:

Understanding Google Cloud Security and Operations

Sub-Objective:

Explain how IT operations need to adapt to thrive in the cloud

References:

[SRE: A Collection of Best Practices for Production Services](#)

Related Material

Question #20 of 60

Question ID: 1495227

You want to create a cloud-based testing environment that duplicates various different hardware setups so that the testing team can perform various application tests within this isolated environment. Which of these technologies should you use for this requirement?

- A) Virtual machines**
- B) Containers**
- C) Serverless data warehouses**
- D) Physical computers**
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Explanation

This is a use case for virtual machines (VMs). A VM is a virtual computer that runs operating systems and applications. Multiple VMs can run on a single physical computer and access various resources from a hypervisor. A hypervisor is software that creates an abstraction of all the computing resources of the physical computer and provides these resources to VMs running on the computer as required. The VM architecture provides the underlying framework in use for modern cloud computing. Compute Engine is Google's VM service, which offers VMs that can be customized. Some uses of VMs in enterprises include:

- Server consolidation – VMs can act as servers to host other VMs and so reduce sprawl by consolidating multiple computing systems into a single physical machine.
- Migration of workloads – VMs are flexible and portable, which enable swift workload migration for companies.
- Test environment creation – VMs can provide an isolated system that can be used for testing without affecting the surrounding systems.
- Disaster recovery – You can duplicate an existing system as a VM on the cloud, thus adding an additional layer of security to the system.
- DevOps – VMs can be activated, deactivated, and migrated smoothly, which facilitates flexible app development.
- Hybrid systems – VMs help you flexibly integrate existing on-premises legacy systems with a cloud environment to create a hybrid computing environment.

You would not use containers for this scenario. Containers are better suited for creating applications that need to be compartmentalized and made portable across multiple platforms. By compartmentalizing an application using containers, you can then change parts of the application in isolation from the other parts. Containers are software packages that contain application code as well as dependencies, such as runtime environments and libraries, that are required for running an application. This allows containers to virtualize an OS and allow applications to be made portable across multiple systems.

You would not use serverless data warehouses for this scenario. A serverless system like BigQuery would be suitable for a requirement where there needs to be very high levels of scalability for working with very large quantities of data. BigQuery is a Google service that provides an enterprise-level data warehouse that is fully managed and that can be utilized for analytics. It has features like machine learning (ML), business intelligence (BI), and geospatial analytics ready for use. It also has a serverless architecture that allows you to use SQL queries to gain key organizational data. BigQuery has a scalable analysis engine that has a distributed structure and allows you to perform queries on petabytes of data within minutes.

You would not use physical computers for this scenario. This is because procuring physical computers for creating testing environments is not as economical, feasible, or flexible as using a VM based system. A single physical computer has one hardware configuration with a specific CPU, RAM, and storage setup. Using several different VMs, you can have several hardware setups, which makes the process of testing an application across multiple platforms much simpler.

Objective:

Infrastructure & Application Modernization

Sub-Objective:

Learn what modernizing IT infrastructure with Google Cloud means

References:

[Cloud.google.com > What is a Virtual Machine?](#)

Related Material

Question #21 of 60

Question ID: 1495296

One of the video streaming applications your company has released faces issues with sudden massive levels of memory consumption causing performance lags on user devices. Your development team needs to find the exact block of application code that may be causing the anomaly. Which Google Cloud service should they use for this?

- A) Cloud Logging**
- B) Cloud Spanner**
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C) Cloud Armor

D) Cloud Profiler

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Explanation

Your development team should use Cloud Profiler for this scenario. Cloud Profiler is used to find and remove any possible performance issues from your applications currently in production. This is accomplished by Cloud Profiler regularly profiling the consumption of resources across your production applications. Cloud Profiler creates an analysis of the performance of an application by collecting data related to CPU usage and memory allocation. It can link performance information for CPU and memory usage to the source code for an application, which allows you to pinpoint specific components of the application that use up the largest amount of computing resources.

You would not use Cloud Logging for this scenario. Cloud Logging is a log management service that is fully managed. It allows you to store, search, and analyze exabytes of log data, as well as set up alerts and notifications. Cloud Logging has Logs Explorer, which lets you use flexible queries for searching, sorting, and analyzing logs. You can also utilize log buckets to store and process log data based on regional compliance requirements.

You would not use Cloud Armor for this scenario. Cloud Armor is a service that protects websites and software applications from threats like web attacks and distributed denial of service (DDoS) attacks. It offers adaptive protection by detecting and mitigating large numbers of Layer 7 DDoS incidents using a locally trained machine learning (ML) system.

You would not use Cloud Spanner for this scenario. Cloud Spanner is a relational database service that is fully managed and ideal for critical business applications. You would use Cloud Spanner for the provision of a highly scalable database for online transaction processing (OLTP). This database will have high levels of availability and consistency worldwide.

Objective:

Understanding Google Cloud Security and Operations

Sub-Objective:

Identify Google Cloud solutions for cloud resource monitoring and application performance management

References:

[Cloud.google.com > Cloud Profiler](#)

Related Material

Question #22 of 60

Question ID: 1495245

Your company is looking for a serverless platform that is fully managed for running containers individually. What is the most efficient way to write code for an application and deploy it as a container with no additional overhead required?

A) App Engine

B) Cloud Run

C) Cloud Functions

D) Compute Engine

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Explanation

You should use Cloud Run for this scenario. You can use Cloud Run for situations where you need to perform deployment of containerized applications that are written in your preferred coding language. Cloud Run provides support for Hypertext Transfer Protocol (HTTP), Hypertext Transfer Protocol Secure (HTTPS), and websockets. You can use Cloud Run for creating application programming interfaces (APIs), applications for processing data, webhooks, and websites. With Cloud Run, code can be packaged into a container and simply deployed without needing infrastructural scaling. The apps containerized with Cloud Run can be written in Ruby, Python, Go, Java, Node.js, or .NET.

You would not use Cloud Functions for this scenario. Cloud Functions is best suited for when you need to run functional code based on an event or a trigger. Cloud Functions runs your code when events occur from various sources like Pub/Sub, Cloud Storage, and Hypertext Transfer Protocol (HTTP). An example is running a process for video transcoding automatically whenever a video is uploaded to Cloud Storage.

You would not use Compute Engine for this scenario. Compute Engine provides virtual machine (VM) computing that has scalability and flexibility. It allows you to utilize Cloud Tensor Processing Units (TPUs), Graphics Processing Units (GPUs), and CPUs to perform analytics and processing on a large scale using Google's technological infrastructure.

You would not use App Engine for this scenario. App Engine allows you to create and host your applications on the same platforms Google uses for several of its own applications. With App Engine, you can build and deploy applications quickly and manage them easily. It provides scalability without the issues related to backups, patching, and hardware provisioning. App Engine provides support for web application development using programming languages like Python, Ruby, C#, Node.js, Go, and PHP.

Objective:

Infrastructure & Application Modernization

Sub-Objective:

Understand modernizing applications with Google Cloud

References:

[Cloud.google.com > Cloud Run](#)

Related Material

Question #23 of 60

Question ID: 1495930

A company with an on-premises data center needs to understand the benefits of adopting an API-based system. Which of these tasks using APIs can modernize an existing legacy on-premises system?

A) Reducing automation to give developers more control

B) Ensuring all teams perform tasks at the same pace

C) Creating a reliable monolithic application for serving users

D) Splitting applications into individual business services

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Explanation

By using application programming interfaces (APIs), you can split existing applications into individual business services. The process of functional decomposition involves identifying individual components and their related systems inside of existing workloads. Then, you can create a decoupled architecture where workloads are broken into smaller parts and scaled independently. This can help you provide higher availability of services and protection from failures. With APIs for various business services, multiple service users and teams can use these services at their own pace. This system also provides greater visibility into the usage of these services in the company.

Reducing automation to give developers more control is not a benefit of using APIs. One of the main advantages of using APIs is to provide automation that reduces the time users spend doing tasks by having more tasks performed by computers. This boosts productivity and also can provide more accountability with tracking systems that record API calls.

Creating a reliable monolithic application for serving users is not a benefit of using APIs. APIs help move away from a monolithic architecture for application development to a microservices-based architecture. This provides greater stability to the system and ensures higher availability as individual components of the system can run independently, and there is no dependency between them. This ensures that if one component fails, then the other application components are not affected.

Ensuring all teams perform tasks at the same pace is not a benefit of using APIs. APIs can be used for various business services, and multiple service users and teams can use these services at their own pace. This system also provides greater visibility into the usage of these services in the company.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Understand the value of APIs

References:
[Inthecloud.withgoogle.com > CIO Guide to Application Modernization](#)

Related Material

Question #24 of 60

Question ID: 1495934

Your company has embraced a modern application architecture by dividing existing systems into microservices and setting up services management for them. Now, what should you use to manage microservices as APIs?

- A) BigQuery

B) Apigee

C) Firestore

D) Dataprep
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Explanation

You should use Apigee for managing microservices as application programming interfaces (APIs). Apigee API Management works in tandem with Envoy and Lstio by enabling API management within a microservices stack. Lstio is an open source system for managing and connecting microservices securely. Apigee is used to build, secure, scale, and analyze APIs at any location with support for the complete API lifecycle. It provides strong systems of API control and visibility. With Apigee, you can increase the value of your digital assets by monetizing your API-based products.

You would not use Firestore for this scenario. Firestore is a serverless document database which is scalable and fully managed. It enables fast development for applications for the internet, mobile, and the Internet of Things (IoT), which can connect to the database directly. Firestore can be used to create real-time applications by using its native synchronization features.

You would not use BigQuery for this scenario. BigQuery is a Google service that provides an enterprise-level data warehouse that is fully managed and that can be utilized for analytics. It has features like machine learning (ML), business intelligence (BI), and geospatial analytics ready for use and has a serverless architecture that allows you to use SQL queries to gain key organizational data. BigQuery has a scalable analysis engine that has a distributed structure and allows you to perform queries on petabytes of data within minutes.

You would not use Dataprep for this scenario. Dataprep is a data service for exploring, cleaning, and preparing data visually for purposes of ML and analysis. It is an intelligent data service and is serverless. This removes the need for users to deploy or manage infrastructure. It also suggests data transformations that are ideal each time a user interface (UI) input is detected, and this removes the need for a user to write code for these applications. Dataprep also detects any anomalies that may exist in the data, including duplicates and missing value data. This helps a user avoid having to assess the quality of data and proceed directly to exploratory data analysis.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Understand the value of APIs

References:
[Cloud.google.com > Apigee API Management](#)

Related Material

Question #25 of 60

Question ID: 1495931

A company uses custom made software to provide business services for consumers. The company is having trouble related to revealing certain aspects of its third-party applications to customers. How can it prevent the leakage of these vendor-related interfaces to its customers?

- A) Use API endpoints and move the code to the cloud

B) Block all exposed APIs to ensure confidentiality

C) Allow customers to install vendor-specific apps

D) Use a monolithic self-made application with no APIs
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Explanation

The company needs to use application programming interface (API) endpoints and move the code to the cloud. By utilizing endpoints that are vendor-agnostic and moving the entire application to Google Cloud Platform (GCP), you can ensure that any interfaces that are specific to vendors and third parties are not revealed to customers. You can move the application code to GCP using Migrate for Anthos.

The company should not allow customers to install vendor specific apps. You need to maintain a level of abstraction between customers and the vendors that create the applications you use to provide services to customers. You need to ensure that the customers do not have direct access to these applications and instead only reveal the necessary services to them using APIs.

The company should not use a monolithic self-made application with no APIs. APIs help move away from a monolithic architecture to application development and move to a microservices-based architecture. This provides greater stability to the system and ensures higher availability as individual components of the system can run independently, and there is no dependency between them. This ensures that if one component fails, then the other application components are not affected.

The company should not block all exposed APIs to ensure confidentiality. Consumers need access to APIs to perform various business tasks, so blocking access to the APIs would disrupt business operations. Instead, you need to provide a more controlled solution by first creating APIs that are vendor-agnostic. With APIs for various business services, multiple service users and teams can use these services at their own pace. This system also provides greater visibility into the usage of these services in the company.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Understand the value of APIs

References:
[Inthecloud.withgoogle.com > CIO Guide to Application Modernization](#)

Related Material

Question #26 of 60

Question ID: 1495229

You need to find the right cloud infrastructure option for migrating from a legacy system with specific licensing needs and running your workloads. Which Google service should you use when you need to have granular control over infrastructural resources, such as CPU and RAM, on a particular OS?

- A) App Engine

B) Cloud Functions

C) **Compute Engine**

D) Cloud Run
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Explanation

You should use Compute Engine for this scenario. You would use Compute Engine when migrating from a legacy system that has specific OS and licensing needs. Compute Engine provides the use of virtual machines (VMs) where you can specify the exact memory, CPU, disk storage, and graphics processing units (GPUs) you need, along with the exact OS and other applications you need to use.

You would not opt for App Engine in this scenario. App Engine provides a serverless platform that is fully managed for building and hosting full web applications. You can write a web application using a preferred language and then use App Engine to manage the tasks related to application and database scaling as well as performing updates to the application. App Engine is the preferred option for when you need to focus on coding a web application or a backend for a mobile app and leave the tasks of deployment and scaling to Google Cloud.

You would not opt for Cloud Functions in this scenario. Cloud Functions is best suited for when you need to run functional code based on an event or a trigger. Cloud Functions runs your code when events occur from various sources, such as Pub/Sub, Cloud Storage, and Hypertext Transfer Protocol (HTTP). For example: running a process for video transcoding automatically whenever a video is uploaded to Cloud Storage.

You would not opt for Cloud Run in this scenario. You can use Cloud Run for situations where you need to perform deployment of containerized applications that are written in your preferred coding language. Cloud Run provides support for HTTP, Hypertext Transfer Protocol Secure (HTTPS), and websockets. You can use Cloud Run for creating application programming interfaces (APIs), applications for processing data, webhooks, and websites. With Cloud Run, code can be packaged into a container and simply deployed without needing infrastructural scaling. Another related container technology is Google Kubernetes Engine (GKE), which allows you to create and manage Kubernetes clusters. Kubernetes provides an open-source system to automatically manage, deploy, and scale container-based applications. GKE provides detailed control over containers and nodes, enabling you to specify CPU, RAM, storage, GPUs, and network configurations, along with an Autopilot option for situations where there is not enough operational and developer support available.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Learn what modernizing IT infrastructure with Google Cloud means

References:
[Cloud.google.com > Where should I run my stuff? Choosing a Google Cloud compute option](#)

Related Material

Question #27 of 60

Question ID: 1497141

Your development team has limited experience building intelligent systems for image classification based on predefined labels. You need a way to train customized models with the least possible effort so that workloads can be parallelized and less time is spent on manual tasks. Which of these technologies can help you achieve this goal?

- A) Looker

B) Dataproc

C) **AutoML**

D) Cloud Spanner
- ✖

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Explanation

You can use Automated Machine Learning (AutoML) to achieve the necessary objective. AutoML enables you to perform training for high-grade machine learning (ML) models with the least manual effort and knowledge of ML. A user can essentially build a working ML model within a few minutes. Developers can submit their datasets and get to use trained ML models that are built using Google technologies, such as transfer learning and Neural Architecture Search. Neural Architecture Search is Google's technology that provides a way to find, generate, evaluate, and train multiple neural architectures for choosing a solution for a customer automatically. Some key technologies within AutoML include:

- AutoML Natural Language – This allows users to perform categorization of input text into their own predefined labels.
- AutoML Tables – This allows data scientists and developers to build and deploy ML models for structured data at high speeds and scale.
- AutoML Translation – This allows developers to use Google Neural Machine Translation (GNMT) for specific use cases for translation.
- AutoML Video – This allows developers to create ML models for video that are customized and scalable.
- AutoML Vision – This allows developers to train ML models for performing image classification based on custom labels.

The underlying concept behind AutoML involves searching using special algorithms to find solutions that are most optimal for each part of an existing pipeline for ML. This enables a system to be built that automates three main aspects of ML including feature engineering, neural architecture search, and hyperparameter optimization.

You would not use Looker for this scenario. Looker is a platform for business intelligence that allows you to access intelligent data for making better business decisions. It also can be embedded into other applications for providing embedded analytics. Looker can be used to create alerts for proactively sending timely insights related to various business data. It also has dashboards that are interactive, which allow for analyzing data in real-time through a trusted data model.

You would not use Dataproc for this scenario. Dataproc provides a fully managed service that is scalable and allows you to run Presto, Apache Flink, and Apache Spark frameworks. You can perform data lake modernization, Extract, Transform, and Load (ETL) operations, and provide security for data science using Dataproc. It uses Apache Ranger and Kerberos for configuring security.

You would not use Cloud Spanner for this scenario. Cloud Spanner is a relational database service that is fully managed and ideal for critical business applications. You would use Cloud Spanner for provisioning a highly scalable database for online transaction processing (OLTP). This database will have high levels of availability and consistency worldwide.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify Google Cloud's solutions for machine learning and AI

References:
[Cloud.google.com > AutoML](#)

Related Material

Question #28 of 60

Question ID: 1495224

You need to help your company choose an appropriate cloud architecture for its operations. Which of these is an advantage of adopting a hybrid cloud architecture?

- A)** Ending data center costs by migrating infrastructure to a public cloud
- B)** Safeguarding all applications and data from public clouds
- C) Availing services from multiple public cloud service providers**
- D)** Maximizing security of sensitive data through on-premises storage
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Explanation

Maximizing security of sensitive data through on-premises storage is an advantage of adopting a hybrid cloud architecture. By using a hybrid cloud architecture, you can provision resources from a cloud service provider (CSP) and not have to procure the same computing resources for the data center. In a hybrid cloud system, a company uses its own data center along with a public cloud system to achieve its business goals. A hybrid cloud also always contains a private cloud. Some key advantages of adopting a hybrid cloud architecture include greater agility in handling usage bursts, leveraging new and advanced analytics technology, and speeding up time-to-market and reducing cycle times for new applications.

Availing services from multiple public CSPs is not a feature of hybrid cloud architectures but of multi-cloud architectures. A multi-cloud deployment involves using multiple public CSPs where each public cloud service has a different function for the company. This allows a company to use one public cloud for database services and another one for analytics. A company can then use a preferred cloud provider for specific services. Remember that using multiple public CSPs indicates a multi-cloud infrastructure.

Safeguarding all applications and data from public clouds is not a feature of hybrid cloud architectures but of private cloud architectures. In this, all the infrastructure and applications that are part of the cloud or virtualization system are running on systems that are inside a company's on-premises data center. This system does not leverage systems that are a part of a public cloud service offering.

Ending data center costs by migrating infrastructure to a public cloud is not an aspect of hybrid cloud architectures but of public cloud architectures. This infrastructure includes applications that are fully functional in a cloud and have no on-premises portions. This infrastructure option may not be suitable for situations where sensitive data needs to be stored locally and not on a public medium.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Learn what modernizing IT infrastructure with Google Cloud means

References:
[Cloud.google.com > Hybrid and multi-cloud architecture patterns](#)

Related Material

Question #29 of 60

Question ID: 1495294

Your company has a large Google Cloud deployment running multiple applications. It needs to find a way to ensure optimal system health and performance for its applications. Which Google Service should you use to gather relevant metadata, events, and metrics from various components of your applications?

- A) Cloud Monitoring**
- B)** Cloud Spanner
- C)** Cloud Vision
- D)** Cloud Filestore
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Explanation

You should use Cloud Monitoring. Cloud Monitoring can be used to monitor the complete infrastructure of a cloud deployment. It provides a view into the uptime, health, and performance of cloud systems. For this, Cloud Monitoring gathers relevant metadata, events, and metrics from sources that include uptime probes, services, app instrumentation, alerts, notifications, and other components of applications. It allows you to stop issues from happening, and it allows you to identify trends. It also reduces overhead related to monitoring through a service that is integrated and provides metrics, dashboards, and alerts all in one place. Cloud Monitoring provides service level objective (SLO) monitoring and metrics collection for virtual machines (VMs) (using Ops Agent) and Kubernetes (using Managed Service for Prometheus).

You would not use Cloud Vision for this scenario. Cloud Vision is used by developers to understand image content through machine learning (ML) models accessed through an application programming interface (API). It provides image classification into various categories, object and face detection inside images, and identification of printed words inside of images.

You would not use Cloud Spanner for this scenario. Cloud Spanner is a relational database service that is fully managed and ideal for critical business applications. You would use Cloud Spanner for the provision of a highly scalable database for online transaction processing (OLTP). This database will have high levels of availability and consistency worldwide.

You would not use Cloud Filestore for this scenario. You would use this when you need a fully managed and highly available file service that has the capability of storing persistent data for shared workloads. You would use Cloud Filestore for enterprise-grade data applications that need storage with durability and persistence, which can be accessed using a Network File System (NFS).

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Identify Google Cloud solutions for cloud resource monitoring and application performance management

References:
[Cloud.google.com > Cloud Monitoring](#)

Related Material

Question #30 of 60

Question ID: 1495253

You must present Apigee as a Google Cloud service that your company should consider using. Which of these are the key benefits of using Apigee for APIs? (Choose two.)

- A) Cluster management**
- B) Centralized disaster recovery**
- C) Problem diagnosis**
- D) Behavior prediction**
- E) Data lifecycle management**
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Explanation
The key benefits of using Apigee for APIs are that it allows you to perform problem diagnosis and behavior prediction. Apigee lets you monitor application programming interfaces (APIs) using artificial intelligence (AI) powered systems. This ensures your APIs deliver expected performance and maintain high availability for customers and developers. You can also identify any API issues and diagnose them easily using the monitoring systems for Apigee. Additionally, Apigee allows you to use AI with your APIs' historical metadata and perform behavior prediction as well as anomaly detection in real-time. Apigee also helps maintain regulatory compliance.

Cluster management is not a feature of Apigee. It is a feature of Google Kubernetes Engine (GKE). GKE allows you to create and manage Kubernetes clusters. Kubernetes provides an open-source system to automatically manage, deploy, and scale container-based applications. GKE provides detailed control over containers and nodes, enabling you to specify CPU, RAM, storage, GPUs, and network configurations, along with an Autopilot option for situations where there is not enough operational and developer support available.

Data lifecycle management is not a benefit of using Apigee. Cloud Storage can be used for Object Lifecycle Management (OLM), where data can be configured so that data is automatically moved to lower-cost storage classes when the data meets specified criteria, such as a period of time passing, or a new version of the data being created.

Centralized disaster recovery (DR) is not a benefit of using Apigee. Cloud Backup and DR is a service that provides a way to backup workloads from both Google Cloud and on-premises systems for data protection that is consistent as well as centralized. It allows you to have a single system to protect file systems, database instances, and virtual machines (VMs) for both VMware and Compute Engine. Using application-consistent backups that are incremental-forever, Cloud Backup and DR ensure low values for both the recovery point objective (RPO) and recovery time objective (RTO) of a company, ensuring swift recovery of business operations using backups.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Understand the value of APIs

References:
[Cloud.google.com > Apigee API Management](#)

Related Material

Question #31 of 60

Question ID: 1495281

Your company has started an initiative to upgrade its existing video streaming service for its customers. The company needs to ensure that its SLA for the service to its customers is met. You have an SLA of 99% defined for service availability. What should be the value of your SLO?

- A) 99.5%**
- B) 89%**
- C) 99%**
- D) 98.99%**
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Explanation

You should use the value of 99.5% for the service level objective (SLO). This is because you need to keep an SLO value that is higher than the service level agreement (SLA), which helps you get early warnings when there is a danger of not meeting the SLA. Failure to meet a set SLA can incur financial penalties for the company. SLAs are visible to customers that use a service while SLOs and SLIs are internal to the development and operations team. If you do not meet an SLO, then there is a possibility of not meeting the SLA, which can cause issues for the company.

The SLO of 99% is incorrect because this value does not allow you to have an early warning of SLAs not being met. You need to keep the SLO value higher than the SLA, as a rule. The SLO of 99% is the same as the SLA of 99%, which indicates that failure to meet the SLO will mean failure to meet the SLA and indicates that there is no early warning possible for the SLA not being met. You need to specify an SLO that exceeds the SLA. An SLO specifies a definite target for a service, which can be 99.999% uptime. SLOs are defined based on user experience and can address user-centric needs, such as latency.

The SLO of 89% and 98.99% are both incorrect because these SLO values are lower than the SLA value of 99%. This raises an unacceptable situation where SLOs will not meet the SLA. For this reason, you need to ensure that the SLO value exceeds the SLA to create a system where there is no possibility of the SLA not being met.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Explain how IT operations need to adapt to thrive in the cloud

References:
[Services.google.com > Increasing Business Value with Better IT Operations: A guide to Site Reliability Engineering \(SRE\)](#)

Related Material

Question #32 of 60

Question ID: 1495283

You work for a software development company that makes two deployments a day for its workplace collaboration tool, which is heavily in use and growing daily. Your development team has come up with a lot of new features that it wants to release, but the specified error budget for the month has been consumed. How will you roll out future deployments?

- A) Once a day

✖
- B) Once a month

✖
- C) Once a week

✖
- D) Halt deployments

✔

Explanation
You will halt deployments. This is because when new features are added to a service, new bugs may be introduced, which can eat into an existing error budget. There can then come a stage where the error budget for a given period is exhausted, and management will need to halt development of new features until the error budget is restored. For example, when a company that deploys code changes twice a day consumes half its error budget, it needs to reduce code deployments to once a day. When three-fourths of the error budget is consumed, the deployments can be reduced to once a month. When the error budget hits zero for a month, deployments will need to be halted.

The other options are incorrect because you cannot roll out any new deployments when the error budget is consumed for the month. This can cause a failure to meet the service level agreement (SLA) with the service users and cause financial loss for the company. The error budget is the measure of the system's possible downtime, which is the difference between the service level objective (SLO) and the total availability figure of 100%. When an SLO is agreed upon, its corresponding service downtime or error budget is also agreed upon. An SLO specifies a definite target for a service, which can be 99.999% uptime. SLOs are defined based on user experience and can address user-centric needs, such as latency.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Explain how IT operations need to adapt to thrive in the cloud

References:
[Services.google.com > Increasing Business Value with Better IT Operations: A guide to Site Reliability Engineering \(SRE\)](#)

Related Material

Question #33 of 60

Question ID: 1495902

Your company needs to assess its readiness for adopting the Google Cloud platform. For this, you need to assess its cloud readiness. Which theme of the Google Cloud Adoption Framework covers the ability of a company to leverage services that are native to the cloud to reduce overhead in its IT operations?

- A) Learn

✖
- B) Lead

✖
- C) Scale

✔
- D) Secure

✖

Explanation

The correct option is scale. This is the theme that covers the level to which a company uses cloud services, provisions workloads, performs automation, and maintains the applications it needs for its business. This theme covers how a business uses native cloud technologies, like serverless computing and automation, to improve its operations and scalability.

The other options are incorrect as they do not measure a company's depth of involvement in leveraging cloud-native systems to improve its operations. The other options are also a part of the themes of the Google Cloud Adoption Framework (CAF) and address different aspects of the cloud readiness of a company. The CAF is a way for an organization to assess its cloud readiness and find actions to prepare itself for successful cloud adoption. The CAF provides a structure of people, processes, and technology that are involved in a company planning a cloud adoption. There are three components to Google's CAF:

- Three maturity phases
- Four cloud adoption themes
- Seventeen epics

A component of the CAF is the four themes:

- Learn – This measures the learning systems in place for an organization to upskill its IT workforce. It also includes the ability of a company to leverage partners who are skilled in cloud technology to augment its existing workforce.
- Lead – This measures the level of support that the IT workforce receives from the company's leadership for cloud migration. It also covers the level of motivation, collaboration, and cross-functionality of the teams. The team structures, executive sponsorship, and budgeting for cloud projects are also covered in this theme.
- Scale – This measures the depth of usage of services that are native to the cloud for reducing overhead in operations, as well as the level of automation you perform for policies and processes which were operated manually. This theme covers how cloud services and resources are provisioned, capacities of workloads distributed, and updates for applications performed.
- Secure – This measures the level of protection in place for your environment from malicious access using a security system that has multiple layers and is identity-centric. This covers the access control systems in place as well as the security strategies and technologies that you use to protect your services.

Objective:
Introduction to Digital Transformation with Google Cloud

Sub-Objective:
Explain why it is critical for businesses to adopt new technology

References:
[Services.google.com > The Google Cloud Adoption Framework](#)

Related Material

Question #34 of 60

Question ID: 1495228

You need to guide your company toward using more cloud-native services to create greater business value. Which of these are characteristics of serverless computing using the cloud? (Choose two.)

- A) Operational costs are higher

✗
- B) Infrastructure is provisioned by the cloud service provider

✓
- C) Code is run by actual servers

✓
- D) Suitable only for predictable scaling needs

✗
- E) Computing resources need to be managed by users

✗

Explanation

Two characteristics of serverless computing include:

- Infrastructure is provisioned and scaled by the cloud service provider (CSP)
- Code is run by actual servers

Serverless computing is a cloud computing model that allows you to create and execute code without needing to manage the underlying servers or pay for any infrastructure that is unused.

Some serverless Google technologies include:

- Application development – You can use Cloud Functions and App Engine for serverless hosting of applications. You can use Cloud Datastore and Firebase for NoSQL serverless databases.
- Analytics – You can use Cloud Storage and Pub/Sub for serverless data analytics work and Dataflow for processing global scale datasets.
- Microservices – Google serverless systems help develop microservices based systems by bringing down the cost of infrastructure by dynamic scaling. Using fully managed serverless systems also reduces operational costs, which removes the need for companies to perform server management. Also, using serverless systems like Pub/Sub can help avoid synchronous call chains by creating precomputed results asynchronously for microservices.

Computing resources do not need to be managed by users. This is done by the CSP and allows developers to focus on creating business value instead of performing resource provisioning. Also, with the automatic and dynamic nature of resource provisioning using serverless systems, applications developed on serverless systems can perform massive levels of scaling.

Serverless systems do not create higher operational costs. This is because when provisioning servers for database and computing applications is done by the CSP, the operational costs to provision and manage massive server clusters reduce substantially.

Serverless systems are suitable not only for predictable scaling needs but excellent for unpredictable workloads. Serverless systems like Google's Cloud Functions and App Engine both offer dynamic scaling based on request volumes without the need for performing server management by the developer. You can use Cloud Functions for executing small chunks of code triggered by Hypertext Transfer Protocol (HTTP) requests or cloud events. Since the billing is only done per request, this can result in significant savings for traffic patterns that are uneven or unpredictable. Similarly, you can use App Engine for larger sections of code like for a Website.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Learn what modernizing IT infrastructure with Google Cloud means

References:
[Cloud.google.com > Serverless Architecture](#)

Related Material

Question #35 of 60

Question ID: 1495935

A banking company has started using APIs with its existing teller terminals for providing instant message notifications to customers about their latest transactions. The company now wants to measure how much this technological step is improving its operations. What would be a key advantage of using Apigee in this scenario?

- A) Connecting the teller machines to the internet via APIs

✗
- B) Usage of dashboards to view metrics related to APIs

✓
- C) Allowing users to message other banking users using APIs

✗
- D) Letting the APIs change the location of the teller machines

✗

Explanation

Usage of dashboards to view metrics related to application programming interfaces (APIs) is a key advantage of using Apigee in this scenario. Apigee lets you monitor APIs using artificial intelligence (AI) powered systems. This ensures your APIs are delivering expected performance and maintaining high availability for customers and developers. You can also identify any API issues and diagnose them easily using the monitoring systems for Apigee. Additionally, Apigee allows you to use AI with your APIs’ historical metadata and perform behavior prediction as well as anomaly detection in real-time. Apigee also helps maintain regulatory compliance.

Connecting the teller machines to the Internet via APIs is not an advantage and can pose a significant security risk to the company and its customers. The teller machine network needs to be fully secured from any possible threat actors who may illegally access systems using the Internet.

Allowing users to message other banking users using APIs is not an advantage of using Apigee, and it is not a functionality that needs to be implemented. A user’s transactions and activities must only be permitted within the boundaries of the user’s account with the bank. The privacy of both the user and its data need to be fully secured.

Letting the APIs change the location of the teller machines is not an advantage of using Apigee. Location changes for any hardware are not done using Apigee. Changing the location of teller machines would involve the bank’s staff responsible for managing the infrastructure of the baking services.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Understand the value of APIs

References:
[Cloud.google.com > Apigee API Management](#)

Related Material

Question #36 of 60

Question ID: 1495897

Your company has a large on-premises data center that has taken a considerable amount of investment to set up. The company is now looking toward growing its business through the use of a hybrid cloud.

Which of these would be a consequence of using a hybrid cloud solution?

- A) Reduced ROI by the addition of a public cloud**
- B) Increased privacy and compliance for applications**
- C) Increased latency for end users**
- D) Reduced maintenance of on-premises servers**

Explanation

You can increase the privacy and compliance of your applications when using a hybrid cloud solution. This is because you can specify the position of your application in relation to the hybrid cloud computing model. This helps you regulate your application, ensuring they fall within compliance requirements and ensure the necessary privacy controls for them.

Reduced maintenance of on-premises servers is not a consequence of using a hybrid cloud solution. While a hybrid cloud solution can allow you to migrate on-premises servers to the cloud and have the cloud provider maintain the hardware for the servers, maintenance for existing on-premises servers will be unaffected by a hybrid cloud solution.

A reduced return on investment (ROI) by the addition of a public cloud does not occur as you increase your computing capabilities through the addition of a public cloud provider to your existing on-premises systems. This can be done without an increase in your existing data center costs leading to an improvement in ROI.

Using a hybrid cloud solution does not increase latency for end users. When you have applications that are distributed at various remote locations, you can greatly benefit from using a hybrid cloud solution. Hybrid computing occurs near the end user’s location and leads to reduced levels of latency.

The three main cloud deployment models you should know about include:

- Cloud-based (also known as Cloud native) – In this deployment, applications are fully functional in a cloud and have no on-premises portions.
- On-premises (also known as private) – In this deployment, all the infrastructure and applications that are part of the cloud or virtualization system are running on systems that are inside a company's on-premises data center.
- Hybrid – In this deployment, infrastructure and applications are shared between an on-premises environment and a cloud environment. A hybrid cloud model is used by companies that want to take advantage of the scalability and security offered through a public cloud while maintaining data on-premises for regulatory compliance needs.

Objective:
Introduction to Digital Transformation with Google Cloud

Sub-Objective:
Explain why cloud technology is revolutionizing business

References:
[Cloud.google.com > What is a Hybrid Cloud? > What are hybrid cloud benefits?](#)

Related Material

Question #37 of 60

Question ID: 1495927

Your company wants to digitally transform its existing systems. Which of these are key business drivers for modernizing applications for your company? (Choose two.)

- A) Increasing system visibility and control over spread-out operations**
- B) Creation of more agile systems at optimized costs**
- C) Increasing the TCO for facilitating necessary growth**
- D) Increase staff productivity through greater change management**
- E) Remove the need for data gathering from changing markets**

Explanation

Increasing system visibility and control over spread-out operations and the creation of more agile systems at optimized costs are both key business drivers for modernizing applications. By modernizing applications and using cloud-native services, you can have granular control over your development and operational systems, which can be automated to ensure a robust, change-friendly system. With fully managed services and serverless technology, you can have more agile systems in place with lower costs due to lessened overhead for the company.

Increasing the total cost of ownership (TCO) for facilitating necessary growth is not a key business driver. The TCO would be significantly reduced when performing application modernization. This is because application modernization will require less investment by the company for leveraging the latest trending technologies as well as the ability to use managed services for greater scalability and agility. Application modernization ensures that less time and resources are spent on tasks like resource provisioning, scaling, backups, and version updating, which cuts down on the capital that would be required to implement similar functionalities on a legacy system.

Removing the need for data gathering from changing markets is not a key business driver. When modernizing applications, you are able to gather more data from both historical sources as well as trending markets using modern data gathering systems and tools for business intelligence (BI). Application modernization will enable you to use fully managed services for creating data lakes and data warehouses for big data analysis and use query services to deliver useful insights on the same for powering business decisions.

Increasing staff productivity through greater change management is not a key business driver. By using managed services and automation, you can ensure that staff members of varying technical skill levels can achieve greater productivity without the need for intensive change management. Managed artificial intelligence (AI) systems can ensure that even people without the necessary coding skills for creating machine learning systems can use cloud services to build systems that deliver business value by leveraging AI technology to better understand markets and customers.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Understand modernizing applications with Google Cloud

References:
[Inthecloud.withgoogle.com > CIO Guide to Application Modernization](#)

Related Material

Question #38 of 60

Question ID: 1497123

Your company is a gaming application development company. It is considering various cloud service models. Which of these would you be able to do with an IaaS cloud deployment?

- A) Running testing and development operations

✓
- B) Using an entire application stack from the CSP

✗
- C) Updating the backend infrastructure for virtual machines

✗
- D) Developing code without managing the software development platform

✗
- E) Performing analysis for big data

✓

Explanation

Using an Infrastructure as a Service (IaaS) model you can run testing and development operations and perform analysis for big data. IaaS is a cloud service model that provides infrastructure resources to users on demand via the Internet. These resources provide computing, storage, virtualization, and networking services and offer scalability. IaaS provides networking and computing resources, which you can use as a suitable environment for running cycles for testing and development. Also, the scalability if IaaS allows you to combine various disaster recovery (DR) systems into a single virtualized system. IaaS can be used for big data analysis as it has the processing capabilities for handling massive workloads.

You would not use an entire application stack from the cloud service provider (CSP) when using IaaS. This is because it is a feature of the Software as a Service (SaaS) model and not of IaaS. A SaaS model provides full applications and the underlying infrastructure, which can be used by a customer through the Internet.

You would not develop code without managing the software development platform on the IaaS model. This is a feature of the Platform as a Service (PaaS) model where the CSP provides all the necessary software and tools you need to create applications and also maintain the software development platform.

You would not update the backend infrastructure for virtual machines when using an IaaS model. This is the responsibility of the CSP and not the customer in an IaaS model. The customer is responsible for managing the OS, data, and applications.

Objective:
Introduction to Digital Transformation with Google Cloud

Sub-Objective:
Explain why cloud technology is revolutionizing business

References:
[Cloud.google.com > What is IaaS?](#)

Related Material

Question #39 of 60

Question ID: 1497135

Your company is in the process of adopting Google AI technology for its business growth. You are evaluating fairness and security in Google's AI and machine learning differentiators. Which of these is a recommended best practice for Google's stipulated responsible AI?

- A) Train models to create automatic inferences based on detected correlations

✗
- B) Conduct short-term fixes to all issues for swift problem solving

✗
- C) Keep all the features in your ML model, including redundant ones

✗
- D) Perform engagement with multiple users in the development process

✓

Explanation

Performing engagement with multiple users in the development process is a recommended best practice for Google's stipulated responsible artificial intelligence (AI) framework. This is part of a best practice of utilizing a design approach that is centered on actual users. The effectiveness of an AI system's predictions and decisions can be gauged from the experiences of actual users. Incorporating inputs from a wide range of users and use cases helps enrich the project with a range of perspectives from users. This eventually benefits the users that actually utilize the technology. The system needs to have enough built-in disclosures to users so that a satisfying user experience can be achieved.

Keeping all the features in your machine learning (ML) model, including redundant ones, is not a recommended best practice for Google's stipulated responsible AI. The recommended practice is to use the ML model, which is simple and meets goals for performance without using redundant features. The raw data used by an ML model needs to be analyzed, and if the model uses sensitive data, then that data should be understood within privacy mandates for the data. The data needs to be checked for errors, bias, and training-serving skews to ensure the optimal performance of the model.

Training models to create automatic inferences based on detected correlations is not a recommended best practice for Google's stipulated responsible AI. You need to ensure that the ML model does not make casual inferences based on correlations that it detects. You also need to be aware of the limitations of the model based on its training data's scope and range. Communicating these limitations to users can help with getting useful feedback about the AI system from users.

Conducting short-term fixes to all issues for swift problem solving is not a recommended best practice for Google's stipulated responsible AI. Short-term fixes may save time but do not offer solutions that are beneficial in the long-term. For this reason, it is important to create a balance between using short-term fixes and long-term solutions. Also, you need to compare the candidate ML model with the deployed version to identify the ways they differ, and to understand how the update will alter the user experience and system performance of the AI application.

Some additional best practices include using metrics like user feedback, false negatives and positives, product health reports, and system performance data for assessing monitoring and training of the ML model. Also, testing the system with both integration tests as well as unit tests to ensure that the performance of the AI system is as expected. This involves studying how the individual components of the ML model work with other components of the application. Also, continuous monitoring of the model needs to be done to ensure the model incorporates feedback from the real-world and user experiences.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify Google Cloud's solutions for machine learning and AI

References:
[Ai.google/responsibilities > Responsible AI practices](#)

Related Material

Question #40 of 60

Question ID: 1495186

An e-commerce company is expanding its workforce and is working on performing data analysis to find ways to grow its business. You need to use a suitable service to improve decision making and productivity in the organization by providing key insights to a growing number of users. Which tool should you use for this requirement?

- A) Dataproc

B) Cloud Trace

C) Dataflow

D) Looker
- ✖

✖

✖

✔

Explanation

You should use Looker. Looker is a platform for business intelligence (BI) that allows you to access intelligent data for making better business decisions. It can be embedded into other applications for providing embedded analytics. Looker can be used to create alerts for proactively sending timely insights related to various business data. It also has dashboards that are interactive, which allow for analyzing data in real-time through a trusted data model.

You would not use Dataproc for this scenario. Dataproc provides a fully managed service that is scalable and allows you to run Presto, Apache Flink, and Apache Spark frameworks. You can perform data lake modernization, Extract, Transform, and Load (ETL) operations, and provide security for data science using Dataproc. It uses Apache Ranger and Kerberos for configuring security.

You would not use Dataflow for this scenario. Dataflow is a fully managed service that automates the process of provisioning and managing processing resources. It provides the highest possible resource utilization by performing horizontal autoscaling for various worker resources.

You would not use Cloud Trace in this scenario as it is not for BI, but a system for tracing that gathers data related to latency from various applications. The Cloud Trace service is useful for identifying any bottlenecks that may exist in your system. The latency data collected by Cloud Trace is displayed on the Google Cloud Console. This lets you track the propagation of requests in your application and view near real-time insights of its performance. Cloud Trace can be used to trace performance data from containers, virtual machines (VMs), and projects for App Engine. It also provides automatic alerts when it detects a deviation from the latency profile of your app.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify common Google Cloud solutions for smart analytics

References:
[Cloud.google.com > Looker](#)

Related Material

Question #41 of 60

Question ID: 1497156

Your company's C-Suite has asked you to take over the task of directing the management of its ongoing Google Cloud expenses. Which of these actions must a company perform to achieve better financial governance of its cloud deployment?

- A) Ensure the on-premises data center matches the cloud for hardware

B) Allow all users across the company to provision and access resources

C) Keep teams siloed for better individual focus on their work

D) Implement throttling on cloud resources exceeding quotas
- ✖

✖

✖

✔

Explanation

A company must implement throttling on cloud resources that exceed quotas to achieve better financial governance of a cloud deployment. Setting resource quotas and performing throttling needs to be done to ensure that cost overruns do not occur due to resource utilization that exceeds the specified budgets. A company can keep various controls when considering a public cloud deployment. These can help ensure better governance over cloud expenditure.

- Specify permissions for cloud resource usage using the principle of least privilege
- Maintain control of cloud resource usage by implementing resource quotas
- Control cloud expenditure using budgetary action automation and notifications

You would not allow all users across the company to provision and access resources. This can cause improper use of cloud resources that leads to wasted capital as well as security issues. Instead, you would opt to use the principle of least privilege for specifying access permissions. The impact of the least privilege principle in access management is that a user only has access to the cloud resources necessary for task completion. This ensures that a user cannot access resources that are outside their scope of work, thus increasing privacy and security on the cloud. This access model ensures that users with malicious intent cannot cause damage to resources outside of their privileges.

You would not ensure the on-premises data center matches the cloud for hardware. The on-premises data center is independent of the cloud deployment in its infrastructure. The purpose of using a cloud deployment is to have a cost-optimized way of scaling computing and storage resources to match business demands without needing to take on the capital expenditure associated with procuring the same level of infrastructure required for a data center.

You would not keep teams siloed for better individual focus on their work. By incorporating a partnership that is implemented across various teams, you will gain greater visibility, which is a necessity for efficient cost management. Reducing budget overruns and unauthorized resource use can be accomplished with more collaboration between IT and finance teams.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Describe financial governance in the cloud and Google Cloud's recommended best practices for effective cloud cost management

References:
[Cloud.google.com > A guide to financial governance in the cloud](#)

Related Material

Question #42 of 60

Question ID: 1497139

Your company uses various CRM systems for user experience management. It needs an intelligent system that can extract value from user data to better understand their needs and sentiments. Which Google technology can you use for this requirement?

- A) Cloud Translation

✖
- B) Memorystore

✖
- C) Looker

✖
- D) Contact Center AI

✔

Explanation

Contact Center Artificial Intelligence (CCAI) is the appropriate choice for this scenario. CCAI is used to enhance the customer service experience for your customers in existing contact centers by leveraging AI technology. CCAI includes Dialogflow Customer Experience Edition (CX), Dialogflow Essentials (ES), Speaker ID, Text-to-Speech, and Speech-to-Text. The CCAI Platform is a Contact Center as a Service (CCaaS) platform that is AI enabled and created on Google Cloud natively. It is built to be used with Customer Relationship Management (CRM) systems and provides a unified source of data for customer experiences. Contact Center AI Insights enables customers to perform value extraction from their data in contact centers. They can use Contact Center AI Insights to search for key information and perform necessary actions based on that data using a console. Customers can also use Contact Center AI Insights to extract topics, key areas, and sentiments from data using advanced analytics.

Cloud Translation is not the appropriate choice for this scenario. Cloud Translation performs the translation of text between languages using Representational State Transfer (RESTful) APIs. You can use Cloud Translation to translate text from English to Spanish to any other language or vice versa. Using Cloud Translation API, you can provide automatic text translation features for an app or Website.

Looker is not the appropriate choice for this scenario. Looker is a platform for business intelligence that allows you to access intelligent data for making better business decisions. It also can be embedded into other applications to provide embedded analytics. Looker creates alerts for proactively sending timely insights related to various business data. It also has dashboards that are interactive, which allow you to analyze data in real-time through a trusted data model.

Memorystore is not the appropriate choice for this scenario. Memorystore is a fully managed service that provides in-memory storage of data for providing fast data access with less than a millisecond of response time. It does this by the deployment of distributed caches for applications. It helps to migrate the cache layer to the cloud without needing to modify existing application code.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify Google Cloud's solutions for machine learning and AI

References:
[Cloud.google.com > Contact Center AI](#)

Related Material

Question #43 of 60

Question ID: 1495941

Your company has various business services in use that require the procurement of third-party software. You need to do a security analysis of this approach. What can be the potential problems that can occur when using third-party software in an existing cloud deployment?

- A) Third-party software lacks the support systems available for cloud-based software

✖
- B) Third-party software reduces a company's innovation goals

✖
- C) Third-party software can be prone to cyberattacks

✔
- D) Third-party software cannot handle important business services

✖

Explanation

Third-party software can be prone to cyberattacks. Third-party software can have security issues in it, which can compromise an organization's data security and make the organization vulnerable to cyberattacks. There needs to be a comprehensive security architecture in place within a company's IT processes to check third-party software and monitor it on a regular basis to ensure there are no security issues.

Third-party software has ample support systems available through various channels that include vendor support, online forums, and support groups. Third-party software can be developed by companies or groups that have a strong grasp of cybersecurity challenges and take the necessary measures and perform development best practices to ensure the software is secure and error free. The security level of software depends greatly on the development methodology employed to create it as well as the level of security awareness of the team that created it. Some open-source frameworks have forums and collaborative groups where open knowledge sharing contributes to mitigating any existing issues.

Third-party software does not reduce a company's innovation goals. The level of innovation that a company can incorporate into its operation depends more on the scale to which it has embraced digital transformation and not on its reliance on third-party software. To reach a state of being fully innovative, an organization will first need to ensure that it has open knowledge sharing, decentralized teams, more autonomous decision making, and use of artificial intelligence (AI) systems by team members of varying technical skill.

Third-party software can handle important business services and serve various critical functions of an organization. The amount of capability available in third-party software varies across industries and vendors. Third-party software can be as powerful as cloud offerings and sometimes even exceed the current capabilities of a cloud software offering. You need to select the right third-party software for the task at hand and make the decision based on various factors, which include the security of the third-party software.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Describe a cloud security approach and Google Cloud security benefits

References:
[Cloud.google.com > Google security overview](#)

Related Material

Question #44 of 60

Question ID: 1497134

You are designing a machine learning (ML) production system for your company. You need to create the ML pipeline in a way that ensures that models meet performance expectations. How would you do this? (Choose two.)

- A) Keep the model validation process at the end of the pipeline

✓
- B) Perform manual updates to ML models post-deployment

✗
- C) Keep a one-time training for the ML model pre-deployment

✗
- D) Keep the data validation process at the beginning of the pipeline

✓
- E) Maintain a tactical maturity phase for ML applications

✗

Explanation

Data validation measures need to be placed at the beginning of the pipeline and model validation needs to be placed at the end. Data validation is used to verify the data types and schemas for the fresh data, which is used for retraining an existing machine learning (ML) model. Model validation checks the performance of an ML model's predictions and ensures that the performance is better than its predecessor.

Keeping a one-time training for the ML model pre-deployment will not help meet performance expectations for the ML model. This is because ML models need continuous training (CT). CT helps validate all the components and code of ML systems and checks existing models and schemas. The purpose of CT is to help ML models adapt to changes in data before the models are deployed. This is done automatically. To automate CT reliably, you need to place validation and quality checks for ML training into the existing ML pipeline.

You would not maintain a tactical maturity phase for ML applications as this would not meet the enterprise-level needs of a ML production system. The tactical maturity phase is when a company is only exploring the possibilities of using artificial intelligence (AI) and ML technology. This maturity level is characterized by the use of limited computing resources like data science virtual machines (VMs) for local data processing and training ML models. The tactical maturity level also involves the exploration of AI possibilities through use cases for proof of value (POV) demonstrations. This maturity level sees controlled costs for ML exploration with hardware and compute resources that are dedicated for use by selected data scientists. The exploratory analysis of data is done using ready-for-use AI systems and tools for data wrangling.

You would not perform manual updates to ML models post-deployment. This is not a suitable approach because manual updating can lead to errors and is not a feasible solution in an enterprise that may have a massive number of deployed ML models that need regular updates of their data as well as the implementation of new ML concepts. An automated pipeline-based approach would be ideal for updating ML models. This approach is used in transformational maturity, which involves orchestration and automation of ML and data pipelines end-to-end.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify Google Cloud's solutions for machine learning and AI

References:
[Services.google.com > Google Cloud's AI Adoption Framework](#)

Related Material

Question #45 of 60

Question ID: 1495289

You want to boost the productivity of your employees by having them perform more high-value work and reduce toil. Which of these activities comprise toil?

- A) Policy-based resource throttling

✗
- B) Manually restarting a service every week

✓
- C) Using the Autopilot mode with GKE

✗
- D) Generating automated log file alerts

✗

Explanation

Manually restarting a service every week is an example of toil. Toil comprises work that may be manual, nonenduring, automatable, repetitive, and tactical. A key goal of Site Reliability Engineering (SRE) is to ensure toil automation takes place. Toil automation removes the need to perform manual, repetitive work, which does not have an enduring value. By using observability tools like Cloud Tracing, Cloud Monitoring, and Cloud Logging, you can gain more insight into services and ensure their correct automation. Creating more automation of services can help reduce toil in an organization, which boosts work efficiency and productivity.

All the other options are examples of cloud-based automation and are not examples of toil.

Generating automated log file alerts can be done using Cloud Logging. Cloud Logging is a log management service that is fully managed. It allows you to store, search, and analyze exabytes of log data, as well as set up alerts and notifications. Cloud Logging has Logs Explorer, which lets you use flexible queries for searching, sorting, and analyzing logs. You can also utilize log buckets for storing and processing log data based on regional compliance requirements.

Using the Autopilot mode with Google Kubernetes Engine (GKE) allows you to create and manage Kubernetes clusters. GKE provides detailed control over containers and nodes, which enables you to specify CPU, RAM, storage, GPUs, and network configs, along with the Autopilot option for situations where there is not enough operational and developer support available.

Setting resource quotas and performing throttling needs to be done to ensure that cost overruns do not occur due to resource utilization that exceeds the specified budgets. Using Google Cloud tools, you can perform automated throttling and capping of resource consumption as well as broadcast budget notification alerts on company communication channels, such as Slack.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Identify Google Cloud solutions for cloud resource monitoring and application performance management

References:
[Increasing Business Value with Better IT Operations: A guide to Site Reliability Engineering \(SRE\)](#)

Related Material

Question #46 of 60

Question ID: 1495940

Your company has multiple teams that have hundreds of employees in them. You need a centralized way to manage user access to cloud resources. Which of these systems would you need to use to authorize which users can access specific Google Cloud resources?

- A) Cloud IAM**
- B) Billing APIs**
- C) Secret Manager**
- D) Encryption at rest**
-
-
-
-

Explanation

You would use Cloud Identity and Access Management (IAM) for this scenario. Cloud IAM enables you to manage cloud resources centrally with granular access control and visibility. With IAM, you can grant access at the resource level, such as providing the role of Subscriber to a user for a Pub/Sub topic. This way, IAM allows you to grant roles in addition to Owner, Editor, and Viewer roles. IAM also provides Recommender that can enable you to detect overly permissive access policies and rightsize them. With IAM, resource access is controlled using contextual attributes, such as IP address or resource type. You can perform IAM policy management and creation using the Google Cloud Console, programmatically using IAM methods, or by command-line access using the gcloud tool. Google Accounts are also supported by IAM. You can utilize Cloud Identity with IAM to synchronize user accounts across multiple projects. IAM also provides workforce identity federation that enables you to perform authentication and authorization of a workforce using external identity providers (IdPs). IAM can provide a full audit history that includes granting, removal, and delegation of permissions for compliance needs.

You would not use Secret Manager for this scenario. You use Secret Manager for storing sensitive data like passwords, application programming interface (API) keys, and certificates securely. With Secret Manager, you have a single source of truth for accessing, managing, and auditing secrets on a Google Cloud deployment. Secret Manager provides lifecycle management with versioning for pinning requests to the newest versions of secrets, as well as the ability to use Cloud Functions to automate the process of rotating secrets. It also allows you to select regions for the replication of your secrets data. Secret Manager is integrated with Cloud IAM for controlling access to secrets.

You would not use encryption at rest in this scenario. Encryption at rest of stored data is a way of implementing security for data stored on the cloud infrastructure. Google utilizes multiple layers of encryption for the protection of data at rest. Google's storage infrastructure encrypts all user data by default before it is stored on physical storage. This way, encryption is performed by Google Cloud at the storage or application layer of the cloud infrastructure. Google has several storage services like Colossus and Cloud Spanner, as well as a key management service.

You would not use billing APIs for this scenario. You can use billing APIs to access various tools created by other vendors to manage project bills programmatically and obtain data on all stock keeping units (SKUs) in their repository and related metadata. This is a way of ensuring clarity for financial governance on a cloud deployment. Other methods include using intelligence recommendations for performing rightsizing of computing resources. This helps ensure that costs are minimized by ensuring that virtual machine (VM) usage is optimized.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Describe a cloud security approach and Google Cloud security benefits

References:
[Cloud.google.com > Cloud Identity and Access Management \(IAM\)](#)

Related Material

Question #47 of 60

Question ID: 1495269

Your company is planning to perform a digital transformation using Google Cloud. Before the transformation process begins, the company needs to know more about Google's security posture for its platform. How does Google implement security for data stored on its cloud infrastructure?

- A) Multi-party authorization**
- B) Encryption at rest**
- C) Encrypting inter-service communication**
- D) Virtual network traffic encryption**
-
-
-
-

Explanation

Google implements encryption at rest to secure data stored on the cloud infrastructure. This is done by utilizing multiple layers of encryption for the protection of data at rest. Google's storage infrastructure encrypts all user data by default before it is stored on physical storage. This way, encryption is performed by Google Cloud at the storage or application layer of the cloud infrastructure. Google has several storage services like Colossus and Cloud Spanner as well as a key management service. Cloud Key Management Service (KMS) is a Google Cloud service that customers can use to manage encryption keys. A related concept is encryption in transit, where Google protects data that is transferred between an on-premises system and Google Cloud or between services. This is done by encrypting the data before it is sent, performing authentication for endpoints, and then decrypting and verifying the data upon arrival at the destination.

Multi-party authorization is not related to the security of stored data but is a way of managing access between services. Using a multi-party authorization system, you can implement a two-person restriction that requires an engineer to get approval from another authorized engineer before performing an action that may be of a sensitive nature. Using this system allows for massive scalability of access management systems for a large number of services that operate on the cloud infrastructure.

Virtual network traffic encryption is not related to the security of stored data, but it is a way of encrypting communication between services. Google encrypts all of the traffic on its virtual networking systems. Additionally, there is authentication performed between services running in the infrastructure, as well as encryption of inter-service data transfer. This layered security approach ensures the security of data even if the network is breached or there are compromises of network devices.

Encrypting inter-service communication is not related to the security of stored data but is a way of ensuring that a service is configured to only receive Remote Procedure Call (RPC) requests from an authorized service. An example is, configuring Google Contacts to allow RPC requests from Gmail. This involves Gmail presenting an end-user permission ticket inside of an RPC request. The ticket ensures the Google Contacts returns data only for the user mentioned in the ticket, which creates a safeguard.

Some fundamental cloud security terms you should know include:

- Privacy – This is ensuring that data that belongs to the company is not shared or leaked to any unauthorized entity inside or outside the company. Privacy controls include access management and authentication systems.
- Availability – This is the ability of a system to offer services as measured as a ratio between total uptime versus downtime. For example, the five 9s of availability indicate that a system is available 99.999% of the time.
- Security – This includes all the technologies and processes in place to ensure that the company's IT systems are safe from unauthorized access or damage. Security systems can range from intrusion detection and prevention systems to multi-factor authentication (MFA) systems for securing access.
- Control – This includes all the systems in place that ensure resources are only accessed by authorized users and in an appropriate manner. Control systems include landing zones that ensure compliance as well as resource policies that ensure safe access to resources.
- Zero Trust – This is a security strategy for cybersecurity where policies for security are enforced using strong user authentication and the principle of least privilege. Zero Trust means that no entity should be trusted, and each actor on the system needs to be verified. Zero Trust systems ensure an improved customer experience, simple network infrastructures, and better defense against cyberthreats.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Describe a cloud security approach and Google Cloud security benefits

References:
[Cloud.google.com > Google Infrastructure Security Design Overview](#)

Related Material

Question #48 of 60

Question ID: 1495929

Your company has an on-premises data center that it uses to provide data processing services to customers. Due to the nature of the legacy system, there have been some business disruptions to the current users of specific services due to network traffic issues, anomalies, and malicious bot attacks.

What would be the best way to solve this issue?

- A) Move apps to the cloud using VMware engine**
- B) Build a document database with Firestore**
- C) Create Kubernetes clusters with GKE**
- D) Use API management with Apigee**

Explanation

You should use application programming interface (API) management with Apigee to solve this issue. API management with Apigee enables you to perform monitoring and analysis of API traffic to help mitigate key problems related to performance. It also allows you to build a layered defense system that includes distributed denial of service (DDoS) attack protection, bot handling, API guarding, as well as web application firewalls (WAFs). Apigee lets you monitor APIs and ensure your APIs are delivering expected performance and maintaining high availability for customers and developers. You can also identify any API issues and diagnose them easily using the monitoring systems for Apigee. API management with Apigee allows you to perform routing based on context, transform payloads, and perform API facades that are user specific. You can start by routing a small portion of the users to the modernized version of the service that uses APIs, and then based on the user feedback and experience, you can route all the users to the API-based modern system.

You would not create Kubernetes clusters with Google Kubernetes Engine (GKE) for this scenario. GKE is not used for API management, which is what this scenario needs. GKE allows you to create and manage Kubernetes clusters. Kubernetes provides an open-source system to automatically manage, deploy, and scale container-based applications. GKE provides detailed control over containers and nodes enabling you to specify CPU, RAM, storage, GPUs, and network configurations along with an Autopilot option for situations where there is not enough operational and developer support available.

You would not build a document database with Firestore. Firestore is a serverless document database which is scalable and fully managed. It enables fast development for applications for the internet, mobile, and Internet of Things (IoT), which can connect to the database directly. Firestore can be used to create real-time applications by using its native synchronization features.

You would not move apps to the cloud using VMware engine. VMware Engine can be used to move applications that are based on VMware onto Google Cloud without any changes to existing applications. It provides all the necessary infrastructure and VMware licenses that allow you to have a software-defined data center (SDDC) on Google Cloud.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Understand the value of APIs

References:
[Cloud.google.com > Apigee API Management](#)

Related Material

Question #49 of 60

Question ID: 1497145

You are in the process of creating a working environment for developers in your various machine learning teams. Which of these options would you use for a development environment that is based on a Jupyter notebook and allows for data processing using a Dataproc cluster?

- A) Firestore**
- B) Vertex AI Workbench**
- C) CCAI**
- D) Dialogflow**

Explanation

You would use Vertex Artificial Intelligence (AI) Workbench for this scenario. This provides a development environment that is based on the Jupyter notebook system and can be utilized for the entire workflow you use for your data science tasks. With Vertex AI Workbench, you can work with data in Dataproc clusters, perform model training, and propagate the results from within the interface of JupyterLab. Vertex AI also contains data labeling jobs that you can use to perform data labeling for datasets that will be used for model training. The Vertex AI Feature Store provides a repository that is fully managed for ingestion, serving, and sharing feature values of ML inside of a company.

Dialogflow is not the appropriate choice for this scenario. Dialogflow provides a suite for the development of conversational apps with voice and text capabilities, which include voicebots as well as chatbots. Dialogflow provides cross-platform connectivity to existing apps for the Internet of Things (IoT), iOS, or Android, and for telephony platforms, including Cisco, Genesys, and Avaya. It also can be used for digital platforms, including Slack, Facebook Messenger, and Google. Dialogflow Essentials (ES) is the standard edition, and Dialogflow Customer Experience Edition (CX) is the advanced edition with features like flow visualizations that are interactive and advanced AI features.

Firestore is not the appropriate choice for this scenario. Firestore is a serverless document database that is scalable and fully managed. It enables fast development for applications for the internet, mobile, and the Internet of Things (IoT) that can connect to the database directly. Firestore can be used to create real-time applications by using its native synchronization features.

Contact Center Artificial Intelligence (CCAI) is not the appropriate choice for this scenario. CCAI is used for enhancing the customer service experience for your customers in existing contact centers by leveraging AI technology. CCAI includes Dialogflow CX, Dialogflow Essentials, Speaker ID, Text-to-Speech, and Speech-to-Text. The CCAI Platform is a Contact Center as a Service (CCaaS) platform that is AI enabled and created on Google Cloud natively. It is built to be used with Customer Relationship Management (CRM) systems and provides a unified source of data for customer experiences. Contact Center AI Insights enables customers to perform value extraction from their data in contact centers. They can use Contact Center AI Insights to search for key information and perform necessary actions based on that data using a console. Customers can also use Contact Center AI Insights for extraction of topics, key areas, and sentiments from data using advanced analytics.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify Google Cloud's solutions for machine learning and AI

References:
[Cloud.google.com > Introduction to Vertex AI](#)

Related Material

Question #50 of 60

Question ID: 1497143

You have an existing ML workflow, and you need to investigate how Vertex AI can be used in it. Which of these stages in your ML workflow can you manage using Vertex AI? (Choose two.)

- A) Asynchronous messaging between applications

✖
- B) ML model training and evaluation

✔
- C) Dataset creation and data upload

✔
- D) System bottleneck identification

✖
- E) Data lake modernization

✖

Explanation

You can manage Dataset creation, data upload, and ML model training and evaluation using Vertex Artificial Intelligence (AI). Vertex AI provides a singular application programming interface (API) platform to build and scale ML models swiftly with features for custom tooling and pre-training. It allows you to manage the AI and ML development lifecycle for a company. These stages of the ML workflow can be managed using Vertex AI:

- Dataset creation and data upload
- ML model training and evaluation
- Hyperparameter tuning
- Uploading and storing of ML models inside Vertex AI
- Deployment of the trained model onto an endpoint for prediction serving
- Sending of prediction requests to the endpoint and setting traffic splits
- Performing management of models and endpoints

You would not use Vertex AI for data lake modernization. You can perform data lake modernization, Extract, Transform, and Load (ETL) operations, and provide security for data science using Dataproc. It uses Apache Ranger and Kerberos for configuring security.

You would not use Vertex AI for system bottleneck identification. You can use Cloud Trace for such applications. The Cloud Trace service is useful for identifying any bottlenecks that may exist in your system. The latency data collected by Cloud Trace is displayed on the Google Cloud Console. This lets you track the propagation of requests in your application and view near real-time insights of its performance.

You would not use Vertex AI for asynchronous messaging between applications. You would use Pub/Sub for such applications. Pub/Sub is a middleware for messaging that enables applications to send messages to each other asynchronously. An application designated as a “publisher” application sends messages to a topic, and all the applications that have subscribed to that topic will receive those messages.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify Google Cloud's solutions for machine learning and AI

References:
[Cloud.google.com > Introduction to Vertex AI](#)

Related Material

Question #51 of 60

Question ID: 1495168

Who is responsible for security in an IaaS cloud service model?

- A) The cloud service provider only

✖
- B) The ISP used by the customer

✖
- C) Both the customer and cloud service provider

✔
- D) The customer only

✖

Explanation

In an Infrastructure as a Service (IaaS) cloud service model, both the customer and cloud service provider (CSP) share the responsibility of security for the cloud deployment. Neither the customer nor the CSP take complete responsibility for the security of the cloud deployment. In a shared responsibility model, the customer is responsible for security in the cloud and the CSP is responsible for security of the cloud. The CSP's responsibility includes the security of the data center, physical servers, and services. The customer's responsibility includes the security of the workloads. To summarize, the following are the responsibility of the CSP:

- Data center security
- Virtualization infrastructure
- Software and hardware infrastructure
- Networking infrastructure
- Global infrastructure, including Availability Zones, Regions, and Edge locations

Customers are responsible for securing all the data that they create and store in the cloud. The following are the responsibility of the customer:

- Patching and configuration of operating systems running on virtual machine instances
- Setup of security groups
- Administration of user accounts and permissions

The Internet service provider (ISP) that provides Internet services to the customer is not responsible for security of a cloud deployment. The ISP only provides the necessary Internet connectivity for access to cloud resources but is not involved with securing either the underlying cloud infrastructure or of applications and data stored on a cloud. Additionally, when connecting to a cloud account over the Internet, security mechanisms like virtual private networks (VPNs) may need to be used for securing data transfer.

Objective:

Introduction to Digital Transformation with Google Cloud

Sub-Objective:

Explain why cloud technology is revolutionizing business

References:

[Cloud.google.com > What is IaaS?](#)

Related Material

Question #52 of 60

Question ID: 1497125

You work for a data sciences company that is exploring cloud-based services for modernizing its data processing systems. You need to find a secure, fully managed service for managing clusters with Google Compute and Kubernetes systems. Which service will you use?

- A) Dataprep

B) BigQuery

C) Dataflow

D) Dataproc
- ✖

✖

✖

✔

Explanation

You will use Dataproc for this scenario. Dataproc provides a fully managed service that is scalable and allows you to run Presto, Apache Flink, and Apache Spark frameworks. You can perform data lake modernization, Extract, Transform, and Load (ETL) operations, and provide security for data science using Dataproc. It uses Apache Ranger and Kerberos for configuring security.

You would not use BigQuery for this scenario. BigQuery is a Google service that provides an enterprise-level data warehouse that is fully managed and that can be utilized for analytics. It has features like machine learning (ML), business intelligence, and geospatial analytics ready for use. It has a serverless architecture that allows you to use SQL queries to gain key organizational data. BigQuery has a scalable analysis engine that has a distributed structure. This enables you to perform queries on petabytes of data within minutes.

You would not use Dataprep for this scenario. Dataprep is a data service for exploring, cleaning, and preparing data visually for the purpose of ML and analysis. It is an intelligent data service and is serverless. This removes the need for users to deploy or manage infrastructure. It also suggests data transformations that are ideal each time a user interface (UI) input is detected, and this removes the need for a user to write code for these applications. Dataprep also detects any anomalies that may exist in the data, including duplicates and missing value data. This helps a user avoid having to assess the quality of data and proceed directly to exploratory data analysis.

You would not use Dataflow for this scenario. Dataflow is a fully managed service that automates the process of provisioning and managing processing resources. It provides the highest possible resource utilization by performing horizontal autoscaling for various worker resources.

Objective:

Innovating with Data and Google Cloud

Sub-Objective:

Identify common Google Cloud solutions for data management

References:

[Cloud.google.com > Dataproc](#)

Related Material

Question #53 of 60

Question ID: 1495225

Your company has various data analytics workloads running on-premises. You are in the process of using a distributed deployment pattern to create a hybrid cloud environment for your company. Which of these actions should you take to create a hybrid cloud environment for analytical workloads?

- A) Perform dataset processing on a private cloud

B) Deploy analytical workloads on a private cloud

C) Use a public cloud for creating a data lake

D) Load raw data from a public cloud to a private cloud
- ✖

✖

✔

✖

Explanation

You should use a public cloud for creating a data lake. Using cloud storage for creating data lakes to be used for analytical processing provides the necessary flexibility for a company to gather petabytes of relevant data efficiently and economically. A data lake allows you to store both structured, semi-structured, and unstructured data from multiple sources in a single place. This data can come from a variety of sources, including multimedia, social media, hard disks, servers, Internet of Things (IoT) devices, clickstream, databases, and transactional information. A data lake provides security and scalability and enables companies to perform ingestion of data from various sources, including edge computing, cloud, or on-premises systems.

You would not deploy analytical workloads on a private cloud because analytical workloads require large levels of data processing with bursting characteristics, which are benefits of public cloud services. Dynamic resource provisioning and scaling offered in public cloud systems ensures efficient processing of massive datasets without the large investments required for implementing similar solutions using a private data center approach.

You would not load raw data from a public cloud to a private cloud. Raw data first needs to be extracted from the workloads that are operating in a private cloud or on-premises system and then loaded for analytics processing onto a public cloud. The results of this processing can then be sent back to existing transactional applications as feedback.

You would not perform dataset processing on a private cloud. This is because a private cloud or an on-premises system does not have the elasticity and scalability required to perform large scale computational and storage tasks that are required for analytics processing. You would instead use a public cloud for processing massive datasets without the need to arrange resources for these tasks.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Learn what modernizing IT infrastructure with Google Cloud means

References:
[Cloud.google.com > Hybrid and multi-cloud architecture patterns](#)

Related Material

Question #54 of 60

Question ID: 1495901

You need to suggest a scaling strategy for a company that wants to move from its maturity level of strategic cloud adoption to one that is transformational. Which of these actions will you suggest for this requirement?

- A) Keep the system immutable with no modifications

✖
- B) Perform each modification when issues are found

✖
- C) Perform each modification using a manual approach

✖
- D) Perform each modification using a programmatic approach

✔

Explanation

You will suggest that the company perform each modification using a programmatic approach. This is because taking a transformational approach involves the use of automation. Automation ensures that changes are made in a constant and consistent manner with low risks to the system. This also ensures that changes can be rolled back, if necessary, to maintain a safe system configuration.

You would not perform each modification using a manual approach. This is not a transformational approach, but a tactical approach that involves changes being made infrequently and can pose high risks for the company. This approach is not recommended.

You would not perform each modification when issues are found. This is not a transformational approach that ensures that risks and problems are mitigated before they actually occur.

You would not keep the system immutable with no modifications. This is because a transformational approach requires rethinking and redefining existing systems and relationships for leveraging new technologies to facilitate operational excellence and business growth.

Objective:
Introduction to Digital Transformation with Google Cloud

Sub-Objective:
Explain why it is critical for businesses to adopt new technology

References:
[Services.google.com > The Google Cloud Adoption Framework](#)

Related Material

Question #55 of 60

Question ID: 1495271

Your company is evaluating the key security benefits of using Google Cloud for its operations. How does Google Cloud allow you to implement preventative controls for ensuring the security of your cloud deployment?

- A) Using third-party libraries

✖
- B) Setting IAM permissions

✔
- C) Setting budgetary alerts

✖
- D) Performing logging and monitoring

✖

Explanation

Setting Identity and Access Management (IAM) permissions is an example of using preventative controls to secure a cloud deployment. You can implement preventative controls using security policies and system architecture. A security policy includes constraints that are implemented programmatically to protect a company and its cloud resources from unauthorized access. The system architecture includes the protection that is available using the existing infrastructure. Preventative controls can include routing configuration, firewalls, subnetting, virtual private clouds (VPCs), and resource hierarchies.

Using third-party libraries is not an example of preventative controls. Third-party libraries are used to provide various functionalities that a development team may require for a project. However, third-party libraries need to be validated for the presence of security issues or bugs that need to be patched. You need to assess how the library sits in with the rest of the development stack. If the library is used for authentication, then it needs more scrutiny than if it is used for the User Interface (UI). Some general guidelines for the use of third-party libraries are to first analyze the scale and usage of the library amongst users globally and gauge their experiences. You also need to know how well the library is managed so future risks can be alleviated.

Performing logging and monitoring is not an example of preventative controls. These are both examples of detective controls. Alerting, analysis, monitoring, and logging are used to identify any malicious activity inside an organization. Integrating Security Information and Event Management (SIEM) systems with Google Cloud Logging is how detective controls can be implemented.

Setting budgetary alerts is not an example of preventative controls. These are controls available to companies for keeping cloud costs and usage under control. These include budgetary alerts, usage quotas, and company permissions. Setting resource quotas and performing throttling needs to be done to ensure that cost overruns do not occur due to resource utilization that exceeds the specified budgets. Using Google Cloud tools, you can perform automated throttling and capping of resource consumption as well as broadcast budget notification alerts on company communication channels, such as Slack.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Describe a cloud security approach and Google Cloud security benefits

References:
[Services.google.com > Google Cloud security foundations guide > Google Cloud foundation security model](#)

Related Material

Question #56 of 60

Question ID: 1495195

A company has an on-premises PostgreSQL server database system and is looking for a cloud-based alternative that is fully managed so the company can remove overhead costs associated with replication, maintenance, and updates. Which of these solutions should you recommend to meet this requirement?

- A) Cloud Data Fusion

✖
- B) Dataproc

✖
- C) Cloud Trace

✖
- D) Cloud SQL

✔

Explanation

You should recommend Cloud SQL for this scenario. Cloud SQL provides a relational database system that is fully managed and works with SQL Server, PostgreSQL, and MySQL. It provides features like flags for configurations, rich extension collections, and ecosystems for developers. Cloud SQL brings down the maintenance costs of databases and provides reliable operations supported by a site reliability engineering (SRE) team that is available around the clock. It has automation built-in for tasks like managing storage capacity and provisioning for databases. It can also be integrated with Google Kubernetes Engine (GKE) and BigQuery while also providing observability of your data using Cloud SQL Insights.

You would not recommend Cloud Trace in this scenario because it is used as a system for tracing that gathers data related to latency from various applications. The Cloud Trace service is useful for identifying any bottlenecks that may exist in your system. The latency data collected by Cloud Trace is displayed on the Google Cloud Console. This lets you track the propagation of requests in your application and view near real-time insights of its performance. Cloud Trace can be used to trace performance data from containers, virtual machines (VMs), and projects for App Engine. It also provides automatic alerts when it detects a deviation from the latency profile of your app.

You would not recommend Cloud Data Fusion in this scenario because it is a fully managed service that provides scalable integration for cloud-native data. It allows users to deploy Extract, Transform, and Load/Extract, Load, and Transform (ETL/ELT) data without requiring coding by using a visual interface. Cloud Data Fusion is created using the Centralized Data Analysis Platform (CDAP) data analytics platform which is open source.

You would not recommend Dataproc in this scenario because it is better suited for applications related to data lakes. It provides a fully managed service that is scalable and allows you to run Presto, Apache Flink, and Apache Spark frameworks. You can perform data lake modernization, perform ETL operations, and provide security for data science using Dataproc. It uses Apache Ranger and Kerberos for configuring security.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify common Google Cloud solutions for smart analytics

References:
[Cloud.google.com > Cloud SQL](#)
[Cloud.google.com > What is Cloud SQL?](#)

Related Material

Question #57 of 60

Question ID: 1497157

Your company's accounting department has raised an issue regarding the high operating costs of the cloud deployment. You are asked to find ways to optimize the costs of running the cloud. What should you do to reduce the Total Cost of Ownership (TCO) using Google Cloud?

- A) Delete idle VM instances

✔
- B) Store data on the most available storage classes

✖
- C) Use larger VM sizes than needed

✖
- D) Utilize historical usage data for monitoring

✖

Explanation

You should reduce the Total Cost of Ownership (TCO) through cost optimization tactics, such as deleting idle virtual machine (VM) instances. Cost optimization involves using managed services, keeping a consumption model, and analyzing expenditure and attributing it correctly. Cloud monitoring dashboards display key system metrics graphically in real-time, offering an updated view of the cloud system. This helps you get a clear analysis of how your systems infrastructure, applications, and services are doing. This way, you can use metrics and logs to track and resolve issues swiftly, thus improving your TCO and reducing Mean Time to Recovery (MTTR). You can also create alarms based on metrics. An example is ensuring that server sizes are selected to match actual computing needs in order to ensure greater cost-effectiveness.

You would not reduce the TCO by using larger VM sizes than needed. Using larger server sizes than required will cause a waste of resources and thus a waste of capital. Instead, you can opt for rightsizing VM instances based on your specific requirements. For this, you can create a custom machine type that allows you to specify the exact CPU and RAM configuration for your specific processing needs.

You would not reduce the TCO by using historical usage data for monitoring. Historical data cannot give you an updated view of a cloud system. Monitoring involves observing important system metrics in real-time using dashboards. This allows you to find ways to optimize costs through more efficient use of cloud resources. An example is monitoring VM instances to identify resources that may be idle.

You would not reduce the TCO by storing data on the most available storage classes. Using the most available storage class is suitable for high availability applications. However, it is also the most expensive, and not all types of data require high availability. By using more suitable storage classes for specific types of data, you can optimize storage costs and save on TCO. Google Cloud Storage offers various storage classes, which include standard, nearline, coldline, and archival. By moving data to less expensive storage classes, such as the archival class, you can save on costs for storing data that is accessed rarely. By using a lifecycle policy, you can automatically ensure that data objects change their storage class based on conditions you specify, which can be the age of the object.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Describe financial governance in the cloud and Google Cloud's recommended best practices for effective cloud cost management

References:
[Cloud.google.com > Best practices for optimizing your cloud costs](#)

Related Material

Question #58 of 60

Question ID: 1497153

You are a consultant for a stock trading company that is setting up its Google Cloud environment. You need to guide the team on some key issues that the company needs to be aware of when managing cloud environments. Which of these concerns should you specify? (Choose two.)

- A) Lack of cost predictability

✓
- B) Security of servers housing virtual machines

✗
- C) More overhead using cloud services

✗
- D) Less cloud usage transparency

✓
- E) Lack of on-premises servers

✗

Explanation

You should specify lack of cost predictability and less cloud usage transparency as key issues the company should be aware of. Lack of cost predictability can be an issue for companies managing cloud environments. A customer application may experience surges in user traffic alternated with periods of low levels of traffic. Due to the unpredictable nature of application usage, it can become difficult for a cloud administrator to specify resource limits. Setting resource limits too low can lead to application performance issues during peak usage, and provisioning more resources than required can lead to wasted capital. Less visibility and transparency in cloud usage leads to the wastage of cloud expenditure due to unutilized resources. By using cloud monitoring tools, a cloud consumer can identify resources that are unused and deprovision them so that they do not contribute to the overall billing. This helps reduce unnecessary costs for a cloud deployment.

The lack of on-premises servers is not a concern when managing a cloud environment because the set-up of an on-premises data center and its operation are not related to a cloud environment. To increase the computing power of a cloud deployment, you can perform scaling either manually or automatically, which provisions more virtual servers when required.

More overhead using cloud services is not a concern when managing a cloud environment. By using fully managed services on Google Cloud, you can reduce your operating costs by removing all the overhead that would be required for various tasks, which include resource provisioning, backups, updates, and patching. This reduction in costs allows you to better utilize your capital to grow your business.

The security of servers housing virtual machines (VMs) is not a concern when managing a cloud environment. This is the responsibility of the cloud service provider (CSP). As per the shared responsibility model of cloud environments, the customer is responsible for securing the data and applications that are hosted on the cloud, and the CSP secures the entire cloud infrastructure, which includes networking systems and physical servers that house VMs.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Describe financial governance in the cloud and Google Cloud's recommended best practices for effective cloud cost management

References:
[Cloud.google.com >A guide to financial governance in the cloud](#)

Related Material

Question #59 of 60

Question ID: 1497147

You are a cloud services consultant for a digital publishing company that provides the news by covering the latest world events through web and mobile apps. You want to find a way for the news to be made available in several local languages across the world automatically. Which of these services can you use for this?

- A) Cloud Translation

✓
- B) Cloud Natural Language API

✗
- C) Cloud Spanner

✗
- D) Cloud Vision

✗

Explanation

You would use Cloud Translation for this scenario. Cloud Translation or Translation Artificial Intelligence (AI) performs translation of text between languages using Representational State Transfer (RESTful) APIs. You can use Cloud Translation to translate text from English to Spanish to any other language or vice versa. By using Cloud Translation API, you can provide automatic text translation features for an app or Website.

You would not use Cloud Vision for this scenario. Cloud Vision is used by developers for understanding image content through machine learning (ML) models accessed through an API. It provides image classification into various categories, object and face detection inside images, and identification of printed words inside of images. Cloud Vision allows you to keep moderation for image content, use analysis of image sentiment for marketing strategies, and create metadata for an image catalog. It also allows you to perform image analysis for images uploaded on requests and can be integrated with Cloud Storage.

You would not use the Cloud Natural Language application programming interface (API) for this scenario. Cloud Natural Language API or Natural Language AI enables you to gain an understanding of natural language using API. This can help you identify the following aspects in a block of text:

- Key entities
- Positive or negative sentiments
- Language used
- Syntax including dependency trees and speech parts

The Cloud Natural Language API can be utilized by developers either through API calls or through reference to a document that may be stored on Cloud Storage.

You would not use Cloud Spanner for this scenario. Cloud Spanner is a relational database service that is fully managed and ideal for critical business applications. You would use Cloud Spanner to provision a highly scalable database for online transaction processing (OLTP). This database will have high levels of availability and consistency worldwide.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify Google Cloud's solutions for machine learning and AI

References:
[Cloud.google.com > Translation AI](#)

Related Material

Question #60 of 60

Question ID: 1497154

Your company's cloud deployment is used by several teams and departments. You need to find an efficient way to ensure that the operating costs of the cloud deployment stays within the set budgetary limit specified by management. Which of these actions should you take to accomplish this?

- A) Use organizational policies for resource access

✖
- B) Set resource quotas and perform throttling

✔
- C) Use cloud resource labels and hierarchies

✖
- D) Manage inventory for on-premises systems

✖

Explanation

You should set resource quotas and perform throttling. Setting resource quotas and performing throttling needs to be done to ensure that cost overruns do not occur due to resource utilization that exceeds the specified budget. Using Google Cloud tools, you can perform automated throttling and capping of resource consumption as well as broadcast budget notification alerts on company communication channels, such as Slack. The most important capabilities for cost management on the cloud include:

- Maintaining costs within budgetary limits
- Finding ways to save on IT costs
- Performing forecasting for IT expenditure
- Maintaining consistency in IT costs and keep deviations limited

You would not use organizational policies for resource access in this scenario. You would use these policies to gain more control over your cloud costs. The purpose of using organizational policies is to specify permissions for users to perform spending on Google Cloud based on their role in the organizational hierarchy. This way, you control the ability of specific users to view costs and perform administrative functions using Cloud Identity and Access Management (Cloud IAM) for cloud resources.

You would not use cloud resource labels and hierarchies in this scenario. You can achieve greater accountability in your cloud deployment for financial governance by using cloud resource labels and hierarchies. Using resource hierarchies simplifies the process of allocating costs to specific departments to ensure accountability. This also provides insights into the return on investment (ROI) of the investment you have made on the cloud.

Managing inventory for on-premises systems is not a way of implementing budgetary control on a cloud deployment. An on-premises system and its inventory are independent of cloud resources. The financial management of on-premises systems is not related to the financial governance systems that need to be implemented to manage costs on a cloud deployment.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Describe financial governance in the cloud and Google Cloud's recommended best practices for effective cloud cost management

References:
[Cloud.google.com > A guide to financial governance in the cloud](#)

Related Material